

Can Remote Personalized Digital Counseling Improve Postpartum Contraceptive Use?*

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Abstract

Postpartum family planning (PPFP) is essential for promoting healthy birth spacing and supporting women’s reproductive autonomy. We evaluate a personalized digital counseling intervention delivered via text messages in a randomized controlled trial of 4,751 pregnant women across 20 counties in Kenya. The intervention—built on an existing government-endorsed digital health platform—combined (i) a shared decision-making tool that ranked contraceptive methods based on user preferences, (ii) behavioral nudges to support planning and follow-through, and (iii) targeted educational messages focused on birth spacing and the lactational amenorrhea method (LAM). Message content and delivery were aligned with national and global PPFP guidelines. The intervention improved LAM knowledge by 36% at 3 months postpartum, increased intention to continue family planning among existing users by 4% at 6 months, and raised perceived counseling quality by 2%. We find no significant effect on modern contraceptive use at either 3 or 6 months postpartum—the study’s primary outcomes. Analysis of mechanisms suggests that limited engagement with the personalized content, persistent demand-side frictions, and supply-side constraints contributed to the null effects. Nevertheless, improvements in knowledge, intentions, and counseling perceptions may yield downstream benefits, particularly given the recurring nature of contraceptive decision-making. These findings highlight both the potential and the limits of light-touch digital interventions to support reproductive health behaviors at scale.

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1 Introduction

Maternal and child mortality remain pressing challenges in many low- and middle-income countries (Cresswell et al., 2025; UN IGME, 2025b). In Kenya, the setting for this study, the maternal mortality ratio stands at approximately 379 deaths per 100,000 live births (World Health Organization, 2025), and the under-five mortality rate is 40 deaths per 1,000 live births (UN IGME, 2025a). These figures remain far above global targets, including the Sustainable Development Goals, which call for fewer than 70 maternal deaths per 100,000 and fewer than 25 under-five deaths per 1,000 by 2030 (United Nations, 2023).

One important, modifiable risk factor for maternal and child mortality is closely spaced pregnancies. The World Health Organization (WHO) recommends a minimum interpregnancy interval of 24 months to reduce the risk of adverse outcomes such as preeclampsia, uterine rupture, preterm birth, and low birth weight (World Health Organization, 2013; King, 2003; Rousso et al., 2002; Conde-Agudelo et al., 2007). Yet in Kenya, nearly half of all pregnancies occur within two years of a previous birth (Moore et al., 2015).

Contraceptive counseling and access to effective postpartum family planning (PPFP) methods are critical for supporting healthy birth spacing and reproductive autonomy. In Kenya, uptake remains suboptimal: only about 50% of women report using a modern method by six months postpartum (Track20 Project, 2021), despite national guidelines and evidence highlighting the importance of PPFP.

Barriers to postpartum contraceptive uptake are multifaceted, beginning with gaps in timely, clear, and personalized information (Blazer and Prata, 2016). The lactational amenorrhea method (LAM) illustrates this challenge: though more than 98% effective when three conditions are met—exclusive breastfeeding, infant under six months, and no return of menses—it remains underused and poorly understood (Labbok et al., 1997; Tran et al., 2018). National and international guidelines recommend transitioning from LAM to another modern method by six months postpartum or earlier if any condition no longer holds (Kenya Ministry of Health, 2018; World Health Organization, 2013). Persistent confusion around LAM’s correct use reflects broader informational shortcomings in postpartum counseling.

In addition to information barriers, systemic and behavioral factors limit contraceptive uptake. Health systems often face staff shortages, high patient volumes, and inconsistent provider training, which reduce the quality of in-person counseling (Tumlinson et al., 2015; Ouedraogo et al., 2020). Physical access and affordability also present challenges. Long distances to facilities and high out-of-pocket costs restrict access to services, particularly to long-acting or permanent methods (Benova et al., 2019; Naanyu et al., 2013). Even when services are available, behavioral constraints such as difficulty planning ahead or acting

on intentions can limit uptake ([McConnell et al., 2018](#)). These barriers persist despite national and global guidelines that emphasize respectful, integrated, and preference-sensitive counseling across the antenatal, delivery, postnatal, and immunization continuum ([World Health Organization, 2013](#); [Kenya Ministry of Health, 2018](#); [Blazer and Prata, 2016](#)).

Digital platforms offer a promising way to expand access to contraceptive counseling in overstretched health systems. In Kenya, where mobile phone penetration is high, short message service (SMS) interventions can deliver timely, personalized information directly to women, offering a potentially scalable and cost-effective complement to in-person care. However, rigorous evidence on the effectiveness of these interventions for postpartum contraception remains limited.

To address this gap, we conducted a randomized controlled trial (RCT) enrolling 4,751 pregnant women across 20 counties in Kenya. The goal of the intervention was to strengthen women’s knowledge of contraceptive methods, address misconceptions about postpartum pregnancy risk and the importance of healthy birth spacing, and support informed method choice, planning, and uptake. All participants were enrolled in PROMPTS, an AI-powered SMS platform developed by Jacaranda Health that delivers personalized maternal and newborn health messages tailored to each woman’s pregnancy and postpartum stage ([Jacaranda Health, 2023](#)). The standard PROMPTS content includes messages on antenatal care, postpartum care, and infant immunizations, as well as six SMS text messages focused on PFP.

Participants randomized to the treatment group received additional enhanced content, which included: (1) a shared decision-making tool that ranked contraceptive methods based on user preferences; (2) behavioral nudges to support method planning and follow-through; and (3) targeted educational messages, with a particular focus on birth spacing and the lactational amenorrhea method (LAM). The intervention was co-developed with local clinical and human-centered design experts, with careful attention to cultural sensitivity and consistency with both national and international family planning guidelines. The goal was not to coerce women into adopting any particular method, nor to replace the role of healthcare providers. Rather, the intervention aimed to support women in understanding the benefits of postpartum contraception and in identifying methods aligned with their fertility goals and personal preferences. Upon forming an initial preference, women were encouraged to discuss their choices with a provider during one of the many routine contact points throughout pregnancy and the postpartum period.

Our primary outcomes were self-reported use of modern contraceptive methods at two key postpartum time points: 3 months (early postpartum) and 6 months (mid-postpartum). These time points align with critical windows for contraceptive initiation, including the typical end of exclusive breastfeeding and the recommended transition from LAM to other

modern methods. Secondary outcomes included uptake of long-acting or permanent methods, satisfaction with the contraceptive method used, knowledge of LAM, and intention to continue family planning among existing users. We also assessed participants' perceptions of the quality of the counseling messages. These outcomes provide insight into how the intervention may influence contraceptive behavior by fostering understanding and supporting better alignment between individual preferences and method choice.

The intervention improved knowledge of LAM by 36% at 3 months and increased intention to continue family planning by 4% at 6 months. It also enhanced user-rated message quality by 2%, primarily due to higher perceived helpfulness of treatment group messages for learning about and selecting contraceptive methods. Despite these improvements, we found no overall impact on modern method uptake at either 3 or 6 months postpartum. These null results are precise. We can rule out increases in uptake larger than 4.3% at endline, compared to the control group's uptake rate of 69%. Heterogeneity analyses along many dimensions show that the effect of the intervention on primary outcomes is consistently negligible across subsamples, including among women who, *ex ante*, were expected to have greater unmet demand for contraception. This includes those who no longer want children, those who are contraceptive naive, and those living within walking distance of their antenatal care clinic.

Exploration of mechanisms suggests two related reasons for the null results on contraceptive uptake. First, interaction with the counseling tool was limited: only 20% of women chose to engage with the shared decision-making tool when prompted, and just 6% completed the full counseling flow. Second, among those not using any method, nearly half cited lack of desire to delay or avoid pregnancy. The remainder cited demand-side reasons such as fear of side effects, infertility, or impact on breastfeeding, along with persistent knowledge gaps. These gaps may not have been addressed effectively due to limited engagement with the intervention. Treatment group women were slightly more likely to cite supply-side barriers, though these were mentioned by only 6% of women.

These findings should be interpreted in the context of both the study population and the intervention's delivery model. Participants in this study were drawn from the PROMPTS platform, which tends to attract women with higher levels of education and prior exposure to reproductive health services compared to the general population of pregnant women in Kenya. Women in the control group received substantial educational content through the standard PROMPTS package. Their uptake of modern contraceptives at 6 months postpartum (69%) was markedly higher than national estimates (approximately 50%) ([Track20 Project, 2021](#)), suggesting a limited margin for additional improvement. Even so, the observed gains in knowledge, contraceptive intentions, and perceived counseling quality among treatment group participants are noteworthy, given the low marginal cost of delivering per-

sonalized digital content at scale. These improvements may generate downstream benefits, especially in light of the recurring nature of contraceptive decision-making.

Our study provides new evidence on the effectiveness of a remote digital counseling tool for PPFP informed by behavioral science. We contribute to three strands of literature.

First, we add to the growing evidence on mobile health interventions for contraceptive use in low-resource settings. A systematic review by [Aung et al. \(2020\)](#) found mixed evidence across eight RCTs, with only three showing effectiveness. Key features of successful interventions included push-based content, tailored messages, interactive features, and motivational support. However, many studies faced implementation challenges such as low engagement or content delivery failures. In Kenya, prior RCTs have yielded mixed results. [Johnson et al. \(2017\)](#) found that an SMS “pull” service improved knowledge but not uptake of contraceptives. [Unger et al. \(2018\)](#) and [Harrington et al. \(2019\)](#) found that tailored SMS “push” service improved contraceptive use. Our study contributes to this literature by testing a combined push–pull approach: proactive, interactive messaging guided by behavioral insights, paired with AI-powered decision support.

Second, we contribute to the broader economic literature on demand- and supply-side drivers of contraceptive use. Many interventions focus on financial barriers, such as subsidies, but these often yield limited results. For example, [Dupas et al. \(2025\)](#) found that full subsidies for modern contraception in Burkina Faso did not increase use or reduce fertility, even when combined with interventions to address misperceptions about social norms or child mortality. [Athey et al. \(2023\)](#) similarly found that providing discounts for long-acting methods had no additional impact when combined with structured counseling. In Kenya, [McConnell et al. \(2018\)](#) found that removing user fees did not substantially raise contraceptive use and highlighted the importance of behavioral barriers. These results are consistent with findings from [Miller et al. \(2025\)](#), who show that correcting misbeliefs about pregnancy risk, aligning partner preferences, and addressing concerns about contraceptive effectiveness may matter more than affordability. Our intervention followed this approach by emphasizing healthy spacing, correcting beliefs about postpartum pregnancy risk, and supporting method planning and follow-through.

Third, we contribute to the literature on choice architecture and digital decision tools in reproductive health. Insights from behavioral economics and cognitive psychology suggest that well-structured tools can improve decision quality, especially when choices are complex or unfamiliar ([Deck and Jahedi, 2015](#); [Bordalo et al., 2013](#); [Thaler et al., 2013](#)). Interventions such as the *My Birth Control* app in the U.S. have improved knowledge and counseling satisfaction, although they did not affect method continuation ([Dehlendorf et al., 2017, 2019](#)). Other client-centered approaches like the Population Council’s Balanced Counseling

Strategy (BCS) have shown some impact on postpartum contraceptive use, particularly when husbands were involved (Salmah et al., 2020), although rigorous evidence on decision quality remains limited. Karra et al. (2021) found that tailored counseling shifted women’s stated ideal method but not actual behavior, possibly due to limited exposure to alternative methods or decision deferral.

Among the most promising tools is the provider-facilitated shared decision-making model evaluated by Athey et al. (2023) in Cameroon, which significantly increased uptake of long-acting methods. Notably, they also cross-randomized price discounts and found that discounts had no added effect beyond the shared decision-making tool. Their results suggest that preference-aligned counseling may be more influential than financial incentives. The authors posit that the shared decision-making tool could be adapted for at-home use. Our study tested this possibility directly by deploying a remote, SMS-only version of the tool across diverse Kenyan settings. The lack of an effect on contraceptive uptake highlights the limits of digital counseling alone, especially in the absence of sustained user engagement and robust health system support.

The remainder of the paper is organized as follows. Section 2 describes the study methods. Section 3 presents the main findings. Section 4 discusses these findings. Section 5 concludes.

2 Methods

2.1 Study Setting and Population

PROMPTS is an AI-enabled SMS platform developed by Jacaranda Health that delivers personalized health information and reminders to pregnant women and new mothers in Kenya to support timely care-seeking (Jacaranda Health, 2023; Ochieng’ et al., 2024; Vatsa et al., 2025). Since its launch, PROMPTS has reached over two million women across the country. Messages are sent on behalf of each woman’s county government, reinforcing the credibility and trustworthiness of the service. The platform is offered free of charge to women, with an estimated lifetime operating cost of approximately \$0.74 per user (Jacaranda Health, 2023).

Enrollment typically occurs during a woman’s first antenatal care visit or shortly after delivery and may be initiated either by Jacaranda Health using public facility patient databases or by the woman herself through direct sign-up at the facility. Participation is confirmed via opt-in SMS. Once enrolled, users receive messages aligned with their pregnancy stage and postpartum timeline. These include reminders for antenatal care, postnatal care, and infant immunizations, as well as educational content on pregnancy, newborn care, breastfeeding, and early childhood development. Messaging continues through 12 months

postpartum. Users can also submit health-related questions to an AI-powered helpdesk, which is supported by human agents when needed.¹

This study was embedded within the PROMPTS platform. Between February and April 2024, we screened pregnant women aged 15–49 for eligibility. Eligible participants were 6 to 8.5 months pregnant, had access to an SMS-capable phone, and provided informed consent.

We note that the study was conducted during a period of policy transition in Kenya’s national health financing system. In October 2024, the government implemented a major reform, replacing the National Hospital Insurance Fund (NHIF) with the Social Health Insurance Fund (SHIF) (Bowmans, 2024). This policy shift disrupted the financing of maternity services and reportedly led to increased out-of-pocket costs, particularly in disadvantaged regions such as the Coast (Kenya Ministry of Health, 2024; Government of Kenya, 2024; Mumbi, 2024; Macharia, 2024). These disruptions may have constrained women’s ability to act on their contraceptive intentions, potentially influencing the intervention’s impact. We further examine the role of supply-side barriers in Section 3.6.

2.2 Conceptual Framework

We model postpartum contraceptive use as the outcome of a dynamic, individual-level decision process under uncertainty, informational frictions, and behavioral constraints. In this framework, women choose from a set of contraceptive options based on their preferences, beliefs about method attributes, and the perceived costs of adoption. These choices are made in a context of competing demands and evolving health needs, particularly salient in the postpartum period.

Let woman i choose a method j from a finite set $\mathcal{J}$, including modern methods and non-use. Her expected utility from choosing method j is given by:

$$\hat{U}_{ij} = u_j(p_i) - c_{ij}$$

where:

- $u_j(p_i)$ captures how well method j aligns with woman i ’s preferences p_i ,
- c_{ij} is the perceived cost of adopting method j , including cognitive, logistical, social components, and relational components²,

¹The platform uses natural language processing (NLP) to triage incoming messages based on clinical urgency. Messages flagged as urgent are escalated to a trained clinical helpdesk agent, who ensures that women in need are promptly connected to appropriate care.

²We do not explicitly model partner preferences as a separate agent in the decision process. However, their influence enters the woman’s decision problem as a relational cost.

- and $\hat{U}_{ij}$ is the ex ante expected utility formed prior to the point of action.

Although this decision may be planned during pregnancy, action is typically required in the postpartum period, where time constraints and stress can reduce the salience of earlier intentions. To capture these behavioral frictions, we introduce a quasi-hyperbolic discount factor $\beta_i \in (0, 1]$, which attenuates the weight given to anticipated benefits at the time of action (Laibson, 1997; Mullainathan and Shafir, 2013). The woman’s actual postpartum decision follows:

$$j^* = \arg \max_{j \in \mathcal{J}} \left\{ \beta_i \cdot \hat{U}_{ij} \right\}$$

This framework highlights three key constraints that can lead to suboptimal contraceptive use:

1. **Preference uncertainty.** Women are often presented with a large and complex choice set of contraceptive methods, each with distinct characteristics related to duration, side effects, reversibility, and user control. In such settings, decision-making can be cognitively demanding. Without structured guidance, women may default to familiar methods, defer decision-making, or rely on suboptimal heuristics—such as selecting based on anecdotal advice, availability, or provider suggestion—rather than on informed alignment with their own values and needs (Thaler et al., 2013; Karra et al., 2021). In this context, a structured decision aid such as a shared decision-making tool can play an important role in helping women clarify their preferences and identify suitable options (Athey et al., 2023). By explicitly incorporating user-defined priorities, such tools can improve decision quality and reduce decisional conflict.
2. **Behavioral frictions:** Women may face behavioral barriers such as present bias, forgetfulness, and cognitive overload, particularly during the postpartum period—a time of significant psychological and logistical demands (Mani et al., 2013; Shah et al., 2012; Mullainathan and Shafir, 2013). Behavioral science research shows that time scarcity, stress, and competing priorities during life transitions like new parenthood can undermine follow-through on long-term intentions, including the adoption of contraceptive methods. Even when women have a preferred method in mind, they may not act on it due to limited cognitive bandwidth. In such contexts, timely cues and behavioral nudges can facilitate planning and follow-through on family planning decisions (Thaler and Sunstein, 2008; McConnell et al., 2018; Mullainathan and Shafir, 2013).
3. **Informational frictions:** Using nationally representative survey data from Sub-Saharan Africa, Miller et al. (2025) show that 40% of women underestimate the proba-

bility of pregnancy without contraception, and that most underestimate the efficacy of hormonal methods, reporting rates three to five times lower than their true efficacy for injections and implants. Misperceptions or incomplete knowledge about fertility return, birth spacing, method eligibility, or LAM effectiveness can lead to delays in initiation or inappropriate method use (Tran et al., 2018). In this context, educational messages that reinforce understanding of postpartum fertility and family planning may provide women with timely, actionable information to support method choice and continuation (Blazer and Prata, 2016).

Our intervention was designed to address these constraints through three integrated components. First, a shared decision-making tool helped women identify methods aligned with their preferences, improving expected match quality $u_j(p_i)$. Second, behavioral nudges reduced the cognitive and logistical costs of follow-through c_{ij} by providing timely reminders tied to routine care. Finally, additional counseling messages addressed persistent informational frictions and reinforced key content when women were most likely to act. Together, these components aimed to increase the likelihood that women adopt a method consistent with their preferences by acting on their prior intentions at the right time.

2.3 Intervention Design

Figure 1 summarizes the timeline and key features of the intervention. Participants across 20 Kenyan counties were randomly assigned in a 1:1 ratio to the treatment or control group, with randomization stratified by geographic location (county group)³ and age group⁴ to ensure balance. Figure B1 shows the geographical distribution of the sample.

All participants received PROMPTS’ standard content, including six messages on PPFP and general messages during pregnancy and the postpartum period. The six PPFP messages were sent at 21, 36, 39, 42, and 45 days, and again at 5.5 months postpartum. These messages covered topics such as the benefits of spacing births by at least two years, modern contraceptive methods, and a reminder that fertility can return soon after childbirth. Prior research has found that women who received these family planning messages were nearly twice as likely to use any contraceptive method and more than twice as likely to adopt a LARC method compared to controls (Jones et al., 2020).

While this standard content provides general information, it does not tailor counseling to individual women’s preferences or align fully with recommended best practices for contraceptive counseling outlined in national and international guidelines. These best practices

³Counties were grouped into four categories to avoid small cell sizes: Central Kenya (which includes Nairobi), Western Kenya, Eastern Kenya, and Coastal Kenya.

⁴Age categories were defined as <24, 25–29, and 30+.

emphasize shared decision-making, personalized support, and the integration of contraceptive counseling across routine maternal and child health services—including antenatal care, delivery, postnatal care, and infant immunization visits—to ensure that women are reached at multiple touchpoints ([Kenya Ministry of Health, 2018](#); [World Health Organization, 2013](#)). To address this gap, the treatment group received three additional components designed in collaboration with local clinical experts and human-centered design consultants to enhance personalization, cultural relevance, and informed decision-making⁵:

1. **Shared decision-making tool:** This feature was designed to help women identify a postpartum contraceptive method aligned with their preferences. At 8 to 9 months of pregnancy, women received a text message asking whether they would like help selecting a method. This timing aimed to engage women during antenatal care, while also reaching them close to delivery, when planning for postpartum contraception becomes more immediate. Those who accepted were prompted to rank their two most important attributes in a contraceptive method (e.g., effectiveness, reversibility, duration)⁶. Based on their input, the platform used a decision tree—developed with clinical guidance—to recommend up to three methods that best matched their stated preferences. Women could then read more about each option and were asked to select one to discuss with a provider. For those who had responded “maybe later” during pregnancy, the offer was repeated at 36 days postpartum. This timing was selected to align with the routine 6-week postpartum check, a standard contact during postnatal care. Clinical guidelines recommend that women who are not exclusively breastfeeding initiate a modern contraceptive method by or before this point, as pregnancy can occur as early as 45 days after birth ([Kenya Ministry of Health, 2018](#); [World Health Organization, 2013](#)).
2. **Behavioral nudges:** To support women in bridging intention and action, the platform delivered a series of behavioral nudges timed around these care touchpoints. The first nudge, sent at enrollment, introduced the concept of family planning and provided an overview of available methods. For women who engaged with the shared decision-making feature, a second nudge sent at 8 to 9 months of pregnancy encouraged them to discuss their preferred method with a provider during their final antenatal care

⁵We note that the study was not designed to isolate the individual effects of each of these three components. Our primary analysis evaluates the overall impact of the bundled intervention. Although we report descriptive data on engagement with individual components, we do not attempt to disentangle their relative contributions to the observed outcomes.

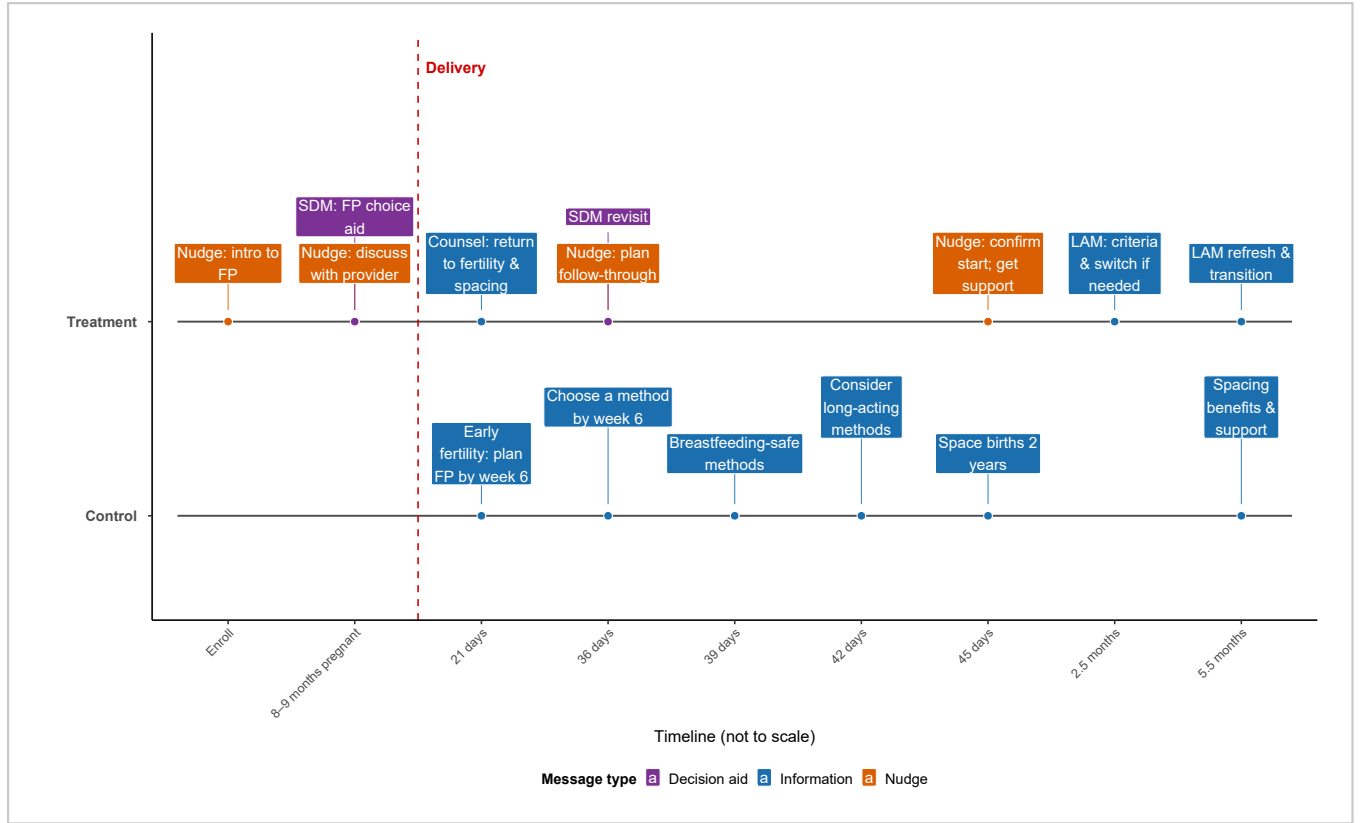
⁶The full list of attribute options included: (1) a method that is highly effective; (2) a method that lasts a long time (3 months or more); (3) a method that allows for regular periods; (4) a method that is safe to use while breastfeeding; and (5) a method that can be stopped when trying to become pregnant soon.

visit or an upcoming postnatal care or child immunization encounter. At 36 days postpartum, a reminder reinforced this plan. At 45 days postpartum, shortly after the recommended 6-week postpartum visit, a message asked whether the woman had started using contraception. If not, she was prompted to indicate the reason, and the system offered tailored support when possible, such as additional information or help locating services.

3. **Additional educational messages:** Women in the treatment group received supplemental counseling messages through 5.5 months postpartum⁷. These messages were designed to address key gaps in postpartum contraceptive knowledge and reflect national and international guidelines recommending early and repeated counseling across the continuum of maternal and child health care. The content included counseling on return to fertility and healthy birth spacing, delivered at 21 days postpartum—well before the 6-week postpartum visit—when many women are still at risk of unintended pregnancy. Two additional message flows were sent at 2.5 and 5.5 months focused on the LAM, a highly effective method if three clinical criteria are met: (i) the baby is under 6 months of age, (ii) the mother is exclusively breastfeeding, and (iii) menstruation has not resumed (Labbok et al., 1997). Women were informed that if any of these conditions were not met, they should transition to another modern method.

⁷Women could also request more information about any method at any time.

Figure 1. Messaging Timeline and Components by Study Arm (Control vs. Treatment)



Abbreviations: FP = family planning; LAM = lactational amenorrhea method; SDM = shared decision-making.

Notes: The figure depicts a discrete (not-to-scale) schedule of PROMPTS SMS content for each study arm. Control group participants received six postpartum *information* messages at 21, 36, 39, 42, and 45 days, and at 5.5 months PP. The Treatment arm received these same six messages plus: (i) a *decision aid* (SDM tool) at 8–9 months of pregnancy, with a revisit prompt at 36 days PP for those who deferred; (ii) a series of *nudges* at enrollment (intro to FP), 8–9 months of pregnancy (discuss plan with provider), 36 days PP (plan follow-through), and 45 days PP (confirm start; offer support); and (iii) additional *information* messages at 21 days, 2.5 months, and 5.5 months PP, including LAM guidance with instructions to transition if LAM criteria are not met. Boxes are color-coded by message type (Information, Decision aid, Nudge). When multiple messages were scheduled on the same day, they were sent as distinct SMS texts.

2.4 Data Collection and Outcomes

2.4.1 Data Collection

Women registered on the PROMPTS platform received a text message notifying them of the opportunity to participate in a research study, with the option to opt out by replying. Those who did not opt out were contacted by trained enumerators for phone-based eligibility screening and informed consent. Eligible and consenting women completed a baseline survey and were then randomized 1:1 to the control or treatment group.

The baseline survey was conducted between February and April 2024, followed by phone-based midline and endline surveys at approximately 3 and 6.5 months postpartum, respectively. Follow-up data collection was timed relative to each woman’s reported delivery date.

The midline survey was initially scheduled based on the expected delivery date collected at baseline. Upon first contact, enumerators asked each woman to report her actual date of delivery. If she was found to be less than 2.5 months postpartum, the call was rescheduled for a later date. The endline survey was then scheduled using actual delivery dates recorded at midline. Enumerators attempted up to eight calls over seven days for each follow-up. All women who completed a survey received mobile airtime worth KES 100 (approximately USD 0.77) as a token of appreciation.

Surveys gathered data on self-reported contraceptive use, method initiation timing, LARC uptake, LAM knowledge, satisfaction with the method used, intention to continue family planning, and perceptions of the SMS content. Enumerators were trained to help respondents approximate timing of method initiation when exact dates were uncertain. At each follow-up, women were reminded that participation was voluntary and that they could skip any questions or withdraw at any point. To reduce emotional burden, women who experienced miscarriage, stillbirth, or infant loss completed a shortened midline questionnaire and were offered information on mental health resources, including the “Nena na Binti” counseling hotline. These participants were not contacted for the endline survey.

Survey instruments were adapted from validated national and international sources, including the Kenya Demographic and Health Survey (KDHS) (KNBS and ICF, 2023), Performance Monitoring for Action (PMA) (PMA Kenya, 2023), and prior family planning studies (Karra et al., 2021; Mandal et al., 2020; McConnell et al., 2018). In addition to demographic and reproductive health modules, the surveys included items on knowledge, attitudes, decision-making autonomy, and partner dynamics. To explore behavioral constraints, we incorporated validated modules on perceived stress, risk preferences, self-efficacy, and executive function, drawing on behavioral economics research conducted in Kenya and globally (Falk et al., 2023; Esopo et al., 2018).

A complete list of prespecified outcomes, definitions, and data collection time points is available on the ClinicalTrials.gov study page⁸. In this paper, we focus on a core set of outcomes most central to the intervention’s theory of change: modern contraceptive use at 3 and 6 months postpartum (primary outcomes), and key secondary outcomes including LARC use, LAM knowledge, method satisfaction, intention to continue family planning, and user-rated quality of the counseling messages.

2.4.2 Primary Outcomes

The primary outcomes were self-reported use of a modern contraceptive method at two time points: 3 months and 6 months postpartum. Modern methods were defined according

⁸<https://clinicaltrials.gov/study/NCT06266780>

to the PMA Kenya classification ([PMA Kenya, 2023](#)), and include sterilization, implants, intrauterine devices (IUDs), injectables, oral contraceptive pills, condoms, diaphragms or cervical caps, spermicides, and LAM.⁹ LAM was only considered a modern method at 3 months postpartum, given that it is no longer effective beyond six months.

We selected these time points based on clinical and policy relevance. The 3-month mark aligns with WHO guidance to initiate contraception by six weeks postpartum ([World Health Organization, 2013](#)), while the 6-month point corresponds with the end of LAM effectiveness and allows measurement of sustained method use.

When participants missed either the midline or endline survey, we used responses from the other survey to impute contraceptive use, following the approach of [McConnell et al. \(2018\)](#). This approach helped address imprecise expected delivery dates reported at baseline. For example, if a woman reported using a modern method at midline and indicated she initiated it before 5.5 months postpartum, we inferred modern method use at 6 months.¹⁰

2.4.3 Secondary Outcomes

We analyzed several secondary outcomes aligned with the intervention’s objectives. First, we measured uptake of long-acting and permanent methods (LAPMs). These comprise long-acting reversible contraceptives (LARCs)—implants and intrauterine devices (IUDs)—as well as sterilization, which represent the most effective contraceptive methods available under typical use ([WHO and Johns Hopkins, 2022](#)).¹¹ These methods provide sustained protection without the need for frequent resupply or strict adherence, making them especially suitable for the postpartum period, when women often face heightened time and resource constraints. Because sterilization was rare in our sample (reported by fewer than 1% of respondents at endline), this outcome primarily captures uptake of LARCs, facilitating comparability with other studies that focus on LARC adoption as a key endpoint. Although injectables and pills are also widely used and effective, they were excluded from this outcome given their shorter duration of protection and higher failure rates ([WHO and Johns Hopkins, 2022](#)).

Knowledge of LAM was a key outcome, measured at the 3-month mark using a three-item index ranging from 0 to 3. The first item assessed whether the respondent had ever heard of LAM. Among those who had, the second item evaluated knowledge of the recommended duration of use (up to 6 months postpartum), and the third captured awareness of the two

⁹Emergency contraception is not classified as a modern method in this study because it is intended for occasional, post-coital use rather than routine pregnancy prevention.

¹⁰Of 4,167 midline respondents, 11 had their 6-month contraceptive use status inferred from midline data. Conversely, 21 of 3,743 endline respondents had their 3-month use status inferred from endline data. Results are robust to excluding these imputed cases.

¹¹LAPMs have failure rates of less than 1% per year under typical use, compared with 7% for pills and 4% for injectables ([WHO and Johns Hopkins, 2022](#)).

additional conditions required for LAM to be effective: the absence of menstruation and exclusive breastfeeding. Participants unfamiliar with LAM automatically scored zero on the second and third items.

Among women who were using a method at follow-up, we also measured two user-reported outcomes related to contraceptive experience, as the intervention was designed to help women identify and select a method aligned with their preferences. These women were asked to rate their satisfaction with their current method using a 1–5 Likert scale, where 1 indicated “very unsatisfied” and 5 indicated “very satisfied,” and to report whether they intended to continue using the method (binary indicator).

Although some of these measures were captured at both follow-up time points, we focus on results at endline, which corresponds to the final time point of data collection. This allows us to capture the most recent information on women’s behaviors and attitudes during the postpartum period. LAM knowledge, however, is reported at midline because the method is only considered effective within the first 6 months postpartum and was not reassessed at endline.

2.4.4 User-Rated Quality of Message Flows

Finally, we assessed participants’ perceptions of the quality of the counseling messages. At 6 months postpartum, participants were asked to rate the helpfulness of the messages using three questions: (1) how well the messages helped them learn about different methods, (2) how well they addressed concerns or questions about contraceptive methods, and (3) how well they supported method selection. Each question was rated from 1 (“not helpful at all”) to 5 (“very helpful”), yielding a total score between 3 and 15. We used this outcome to evaluate how the intervention was received by users and whether it supported informed choice in the postpartum period.

2.5 Analytical Approach

We begin by presenting descriptive statistics on key demographic, socioeconomic, and behavioral characteristics for the full randomized sample, as well as for the analytic subsamples reached at 3 and 6 months postpartum. To test for balance across treatment arms, we compute differences between treatment and control groups using Pearson’s Chi-squared tests for categorical variables and Wilcoxon rank-sum tests for continuous variables (Diez et al., 2019). In addition, we conduct an omnibus test of joint orthogonality (Wald F) by regressing the treatment indicator on the set of baseline covariates and computing a randomization-inference p -value under the actual assignment mechanism (Kerwin et al., 2024).

To estimate program impact, we calculate average treatment effects as risk differences between the treatment and control groups using ordinary least squares. Given the binary nature of most outcomes, we use a linear probability model, which allows for straightforward interpretation of marginal effects in percentage-point terms (Hellevik, 2009). All models include heteroskedasticity-robust standard errors (Wooldridge, 2010).

Our primary specification adjusts for the stratification variables used during randomization (age group and geographic location) to improve statistical precision and account for any residual imbalance (Kahan and Morris, 2012). The estimating equation is as follows:

$$Y_i = \alpha + \beta \cdot \text{Treatment}_i + \boldsymbol{\delta} \cdot \text{Controls}_i + \varepsilon_i \quad (1)$$

where Y_i is the outcome of interest for individual i , Treatment_i is an indicator for assignment to the treatment arm, Controls_i is a vector of categorical indicators for the randomization strata, and ε_i is the error term. The coefficient β captures the average treatment effect on the outcome.

The targeted sample size during study planning was 4,190 women, designed to provide 80% power to detect a minimum detectable effect of 4.25 percentage points, assuming a baseline uptake rate of 57% at 3 months postpartum based on prior evidence (Jones et al., 2020). The realized analytic sample at 3-month follow-up included 4,167 women, yielding a minimum detectable effect of approximately 4.31 percentage points.

In addition to estimating overall treatment effects, we conduct subgroup analyses to explore heterogeneity in treatment effects across dimensions where differential responsiveness is plausible. These include geographic location, baseline fertility intention, intention to use postpartum contraception, and distance to antenatal care facility.

2.6 Ethical Considerations

The study received ethical approval from several oversight bodies. In Kenya, the study was reviewed and approved by Institutional Ethics and Review Committees (IERCs) and was granted a research license by the National Commission for Science, Technology, and Innovation (NACOSTI). It was also approved by the Institutional Review Boards (IRBs) at the Harvard T. H. Chan School of Public Health and the Johns Hopkins Bloomberg School of Public Health, as well as the Ethical and Scientific Review Committee of Amref Health Africa (AMREF) in Nairobi. The trial was registered on ClinicalTrials.gov with the identifier NCT06266780.

3 Results

3.1 Study Enrollment

Figure 2 summarizes participant recruitment and follow-up. Between February and April 2024, 17,466 women enrolled in PROMPTS were screened. Of these, 5,469 (31%) met the eligibility criteria: being 6 to 8.5 months pregnant, aged 15 to 49, possessing an SMS-capable phone, and providing informed consent. The most common reasons for ineligibility were being unreachable by phone or having already delivered. Among eligible women, 4,751 (87%) consented to participate and were randomized 1:1 to the treatment ($n = 2,376$) or control ($n = 2,375$) group.

Midline follow-up at approximately 3 months postpartum was completed by 4,167 women (88%), and endline follow-up at 6.5 months postpartum was completed by 3,743 women (79%).¹² Most attrition was due to repeated failed call attempts, switched-off phones, or call disconnections.

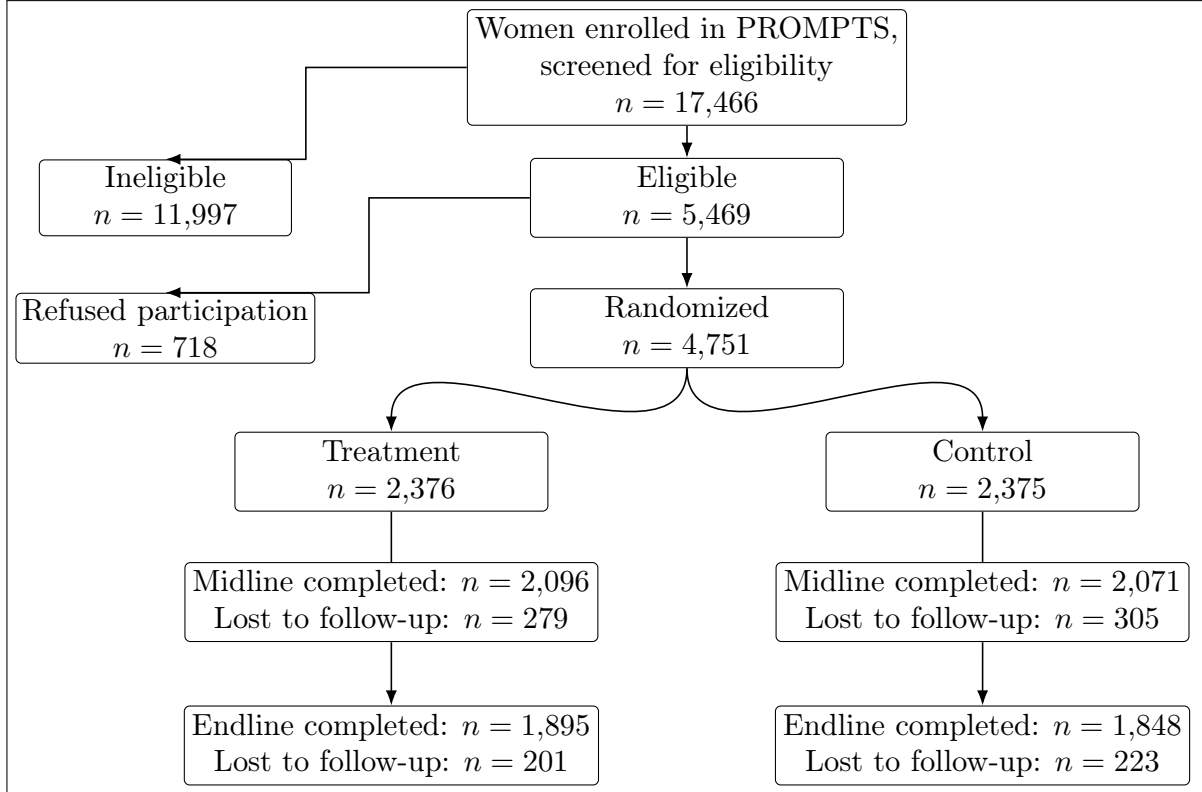
For the 3-month outcomes, we excluded 158 women who reported a stillbirth, miscarriage, or newborn death, and 13 women interviewed before 2.5 months postpartum. For the 6-month outcomes, we excluded eight women interviewed before 5.5 months postpartum and five who reported initiating family planning before delivery. About 25% of endline interviews occurred between 5.5 and 6.0 months postpartum; of these, 99% were between 5.8 and 6.0 months, which we consider sufficiently close to 6 months.

For each outcome, we excluded participants who refused to answer, were unsure, or responded “don’t know”¹³, except for knowledge-based outcomes (e.g., LAM knowledge), where “don’t know” responses were coded as lack of knowledge.

¹²The average (SD) time to midline and endline interviews was 3.5 (0.8) months and 6.5 (0.7) months, respectively.

¹³This was rare. For instance, fewer than 0.1% of participants responded “don’t know” for the primary outcomes.

Figure 2. Flowchart of Participant Recruitment and Follow-up in the Study



Notes: This figure shows the flow of participants from recruitment to endline. Women were screened from the PROMPTS registry between February and April 2024. Eligibility required being 15 to 49 years old, 6 to 8.5 months pregnant, reachable by phone, and able to provide informed consent. Follow-up surveys were conducted by phone at approximately 3 and 6.5 months postpartum. Reasons for loss to follow-up included unanswered calls, disconnected lines, and phones being switched off.

3.2 Descriptive Statistics

3.2.1 Study Sample versus National Survey Subsample

Table 1 presents baseline characteristics of study participants alongside those of a subsample of pregnant women from the 2022 PMA survey in Kenya.¹⁴

The mean age of participants in our study was 26 years, nearly identical to the PMA subsample. A large majority of women were married (79%), and approximately 80% had completed at least primary education. Tertiary education was more common in our sample than in PMA (30% vs. 15%). Geographic distribution also differed: 35% of our sample resided in Central Kenya compared to 24% in PMA, while fewer resided in Coastal Kenya

¹⁴We use PMA data because it is nationally representative with a focus on reproductive health and family planning, and because our instrument was adapted from the PMA questionnaire. The PMA sample shown here is restricted to women who were pregnant at the time of interview. Since pregnancy is not a primary sampling domain, this subgroup is relatively small.

(7% vs. 18%).

On average, women in our study had 1.78 living children at baseline. Roughly 29% reported not wanting additional children, comparable to the 27% in PMA. About one-third of pregnancies were reported as unintended (34% vs. 39% in PMA). Most women intended to use family planning after delivery (85%), slightly below the PMA estimate (87%). Among married women, 93% expected partner support for postpartum family planning (PPFP), compared with 90% in PMA. Prior FP use was also more common in our sample (63% vs. 48%). However, while nearly all PMA respondents reported having a specific method in mind (98%), the proportion was lower in our study (69%). By contrast, women in our study were more likely to plan initiation of FP within six months postpartum (82% vs. 51%).

Two main insights emerge. First, the descriptive statistics indicate substantial demand for PPFP information and services. A notable share of pregnancies was unintended, nearly one-third of women did not want additional children, and the overwhelming majority reported plans to use contraception soon after delivery, often with partner support. These findings point to an opportunity to strengthen informed postpartum contraceptive decision-making.

Second, our study sample appears more socioeconomically advantaged than the nationally representative PMA sample. This difference has implications for external validity, which we discuss further in [Section 4](#).

Table 1. Characteristics of the Baseline Study Sample vs. PMA Sample

Characteristic	Baseline N = 4,751	PMA Sample N = 484
Demographic and Socioeconomic Characteristics		
Age (years)	26.18 (5.78)	26.15 (5.98)
Aged 15–19 (adolescent)	9.2%	12%
Married	79%	73%
Cohabiting, unmarried	1.2%	11%
Highest education: Tertiary	30%	15%
Highest education: Secondary	49%	31%
Highest education: Primary	20%	46%
Highest education: None	0.8%	7.0%
Location: Western Kenya	43%	42%
Location: Central Kenya	35%	24%
Location: Eastern Kenya	15%	16%
Location: Coastal Kenya	6.9%	18%
Ever given birth (live or stillbirth)	57%	74%
Number of living children	1.78 (1.07)	–
FP Knowledge and Beliefs		
Aware of return to fertility postpartum	35%	–
Knowledge index: LAM	0.50 (0.76)	–
Husband supportive of PPFP	93%	90%
Behavioral and Psychological Factors		
Needs reminders to start tasks	13%	–
Difficulty planning ahead	20%	–
Difficulty following through	30%	–
Felt more strained than usual (past month)	31%	–
Access and Financial Constraints		
Travel time to ANC facility (min)	29.46 (23.00)	–
Obtaining 2,000 KSh for care: difficult	0.73 (0.44)	–
FP History and Intentions		
Current pregnancy was unplanned	34%	39%
Does not want more children	29%	27%
Wants to wait 2+ years before next pregnancy	97%	97%
Prior FP use	63%	48%
Last method used was modern	60%	–
Last FP source: public facility	34%	–
Last FP source: private facility	11%	–
Last FP source: pharmacy	18%	–
Intends to use PPFP	85%	87%
Plans to use a modern method postpartum	69%	98%
Plans to start method within 6 months postpartum	68%	–
Weeks to EDD	7.58 (3.31)	–

Abbreviations: PMA = Performance Monitoring for Action; FP = family planning; LAM = lactational amenorrhea method; PPFP = postpartum family planning; ANC = antenatal care; KSh = Kenyan shillings; EDD = expected delivery date.

Notes: This table summarizes characteristics of participants at baseline. Means and standard deviations (SD) are reported for continuous variables; frequencies and percentages are shown for categorical variables. Missing responses are excluded from variable-specific denominators. For comparison, the table also includes summary statistics from the 2022 Kenya round of the PMA survey for pregnant women. A dash (“–”) indicates that a corresponding measure was not available in the PMA dataset.

3.2.2 Randomization Balance

Table 2 reports baseline characteristics overall and by treatment arm. Across most demographic, socioeconomic, and reproductive health measures, the two groups appear similar. Out of the full set of comparisons, only two variables differ at conventional significance levels: knowledge of the LAM ($p = 0.032$) and whether the respondent reported having a modern postpartum method in mind ($p = 0.012$). Given the large number of baseline characteristics examined, a small number of significant differences at conventional thresholds could occur by chance. An omnibus F -test of joint orthogonality yields a p -value of 0.34, indicating that we cannot reject the null of overall balance across arms. Though, to alleviate potential concerns with these two imbalances we include baseline LAM knowledge as a covariate in our regression specifications and verify that the estimated treatment effects are robust to its inclusion or exclusion and, in the heterogeneity analyses, compare treatment effects by baseline intention to initiate PPFP.

Follow-up samples (Tables A1 and A2) show similar distributions across arms and closely resemble the baseline composition, reducing concerns about differential attrition.

Table 2. Baseline Summary Statistics

Characteristic	Overall N = 4,751	Control N = 2,374	Treatment N = 2,377	p-value
Demographic and Socioeconomic Characteristics				
Age (years)	26.18 (5.78)	26.16 (5.79)	26.20 (5.77)	0.8
Aged 15–19 (adolescent)	9.2%	9.1%	9.3%	0.8
Married	79%	79%	79%	0.9
Cohabiting, unmarried	1.2%	1.1%	1.4%	0.4
Highest education: Tertiary	30%	30%	30%	>0.9
Highest education: Secondary	49%	49%	50%	0.6
Highest education: Primary	20%	20%	19%	0.5
Highest education: None	0.8%	0.8%	0.9%	0.8
Location: Western Kenya	43%	43%	43%	0.9
Location: Central Kenya	35%	35%	35%	>0.9
Location: Eastern Kenya	15%	15%	15%	>0.9
Location: Coastal Kenya	6.9%	7.0%	6.9%	0.9
Ever given birth (live or stillbirth)	57%	57%	58%	0.3
Number of living children	1.78 (1.07)	1.81 (1.11)	1.75 (1.04)	0.3
FP Knowledge and Beliefs				
Aware of return to fertility postpartum	35%	34%	36%	0.2
Knowledge index: LAM	0.50 (0.76)	0.48 (0.74)	0.53 (0.77)	0.032
Husband supportive of PPFP	93%	93%	94%	0.6
Behavioral and Psychological Factors				
Needs reminders to start tasks	13%	13%	12%	0.8
Difficulty planning ahead	20%	19%	21%	0.3
Difficulty following through	30%	29%	30%	0.3
Felt more strained than usual (past month)	31%	31%	31%	>0.9
Access and Financial Constraints				
Travel time to ANC facility (min)	29.46 (23.00)	29.07 (22.83)	29.86 (23.16)	0.2
Obtaining 2,000 KSh for care: difficult	0.73 (0.44)	0.73 (0.44)	0.73 (0.45)	0.5
FP History and Intentions				
Current pregnancy was unplanned	34%	34%	34%	>0.9
Does not want more children	29%	28%	30%	0.2
Wants to wait 2+ years before next pregnancy	97%	96%	97%	0.7
Prior FP use	63%	63%	64%	0.7
Last method used was modern	60%	60%	61%	0.6
Last FP source: public facility	34%	34%	35%	0.3
Last FP source: private facility	11%	10%	12%	0.2
Last FP source: pharmacy	18%	19%	18%	0.2
Intends to use PPFP	85%	84%	85%	0.6
Plans to use a modern method postpartum	69%	68%	71%	0.012
Plans to start method within 6 months postpartum	68%	68%	68%	0.7
Weeks to EDD	7.58 (3.31)	7.57 (3.29)	7.60 (3.33)	>0.9
Test of joint orthogonality, p-value				0.3

Abbreviations: PMA = Performance Monitoring for Action; FP = family planning; LAM = lactational amenorrhea method; PPFP = postpartum family planning; ANC = antenatal care; KSh = Kenyan shillings; EDD = expected delivery date.

Notes: This table presents baseline characteristics of participants, overall and stratified by treatment assignment. Means and standard deviations (SD) are shown for continuous variables; frequencies and percentages for categorical variables. P-values are from chi-square tests for categorical variables and Wilcoxon rank-sum tests for continuous variables, testing for differences between treatment and control groups. The joint F test p -value, based on randomization inference, is reported as an overall assessment of baseline covariate balance. Missing values are excluded from variable-specific denominators. Randomization was stratified by age group and geographic location.

3.3 Descriptive Statistics at Follow-up

Tables A3 and A4 present descriptive statistics on pregnancy outcomes, health system contacts, family planning counseling coverage, and exposure to the intervention at midline and endline. At midline, 96% of enrolled women had delivered and had a living infant, 2.5% had experienced infant loss after delivery, and 1.3% had lost a baby during pregnancy.¹⁵

Although only 1% of women reported using the LAM at midline, 36% met all behavioral criteria for effective use: exclusive breastfeeding, no return of menstruation, and infant under six months of age with no other liquids or foods introduced. The gap between implicit and reported use suggests that many women may have practiced LAM unknowingly, highlighting broader gaps in knowledge and labeling. For comparability with national surveys such as PMA, we use self-reported LAM in the main analysis. Nonetheless, the discrepancy underscores the need to interpret early postpartum contraceptive uptake measures with caution.

Contact with the healthcare system during the peripartum period was widespread, but family planning counseling coverage was incomplete and fell short of global and national guidelines. The World Health Organization recommends integrating counseling into all maternal and newborn health contacts, including antenatal care, delivery, postpartum checkups, and child immunization visits (World Health Organization, 2013). Kenya’s national guidelines echo this recommendation (Kenya Ministry of Health, 2018).

At midline, 59% of women reported discussing family planning during antenatal care, 55% during delivery care, and 65% during a child immunization visit. Among those who attended a postpartum checkup—an average of 1.3 visits by midline—71% reported receiving counseling. These figures suggest that while most women received counseling at least once, few were exposed to the repeated reinforcement recommended by guidelines.

At endline, women reported an average of one postpartum visit between midline and endline, and 77% of those visits included counseling. Counseling during immunization visits declined to 56%, compared to 65% at midline. While 92% of women reported discussing family planning at least once across the four recommended contact points (antenatal care, delivery, postpartum checkup, immunization), only 11% received counseling at all four. This inconsistency may limit informed method choice and uptake, particularly for long-acting methods that require provider engagement.

Exposure to the PROMPTS message flows was high. At endline, 98% of women reported still receiving PROMPTS messages. Of these, 81% reported always reading the messages and an additional 14% reported reading them often. High engagement with the messages indicates that SMS-based counseling tools such as PROMPTS can complement in-person

¹⁵At the 3-month interview, no women reported a new pregnancy. By six months postpartum, only 11 women (0.3%) reported becoming pregnant again.

services and help mitigate missed opportunities for family planning counseling.

3.4 User Engagement with Treatment Group Messages

Before turning to the main treatment effects, we summarize engagement with the intervention messages among women assigned to the treatment arm.

3.4.1 Initial Engagement During Pregnancy

Table A5 reports engagement with the initial nudge messages sent at approximately 7.5 months gestation. Of the 2,376 women in the treatment group, 463 (20%) responded to the introductory prompt about PPFP. Among those who engaged, the most commonly requested method was the copper IUD (35%), followed by the lactational amenorrhea method (LAM, 17%) and the Depo-Provera injection (17%). Smaller shares expressed interest in the implant (16%), rhythm beads (11%), the hormonal IUD (11%), the traditional pill (10%), the mini pill (6.5%), and condoms (4.8%). Respondents who indicated interest in one or more methods received tailored informational messages on their selected options.

3.4.2 Engagement with the shared decision-making Counseling Flow

Table A6 summarizes engagement with the shared decision-making counseling flow, delivered at approximately eight months gestation. Of the 2,123 women who received an invitation, 423 (20%) responded by selecting their most valued contraceptive attributes. The most frequently chosen attributes were safety while breastfeeding (36%), the ability to discontinue the method to conceive in the future (30%), and high effectiveness (27%). Based on these preferences, 322 women (15% of those invited) received tailored method suggestions. The most commonly recommended methods were the implant (27%), injectable (22%), and mini pill (17%). A total of 135 women (6% of those invited) ultimately selected a preferred method to discuss with a provider, most often the implant (37%) or injectable (19%).

3.4.3 Postpartum Message Engagement

Table A7 reports engagement with postpartum message flows. At 21 days postpartum, 11% of participants responded to a message inviting them to learn more about specific topics. Among these, 36% selected family planning methods, while others sought information about resuming sex (23%), method timing (19%), and safety (14%).

At 36 days postpartum, 6.5% of participants requested additional information about contraceptive methods. The most frequently selected were injectables (27%) and implants

(20%), followed by the copper IUD (12%), LAM (9.9%), pills (9.9%), condoms (5.4%), and the hormonal IUD (2.7%).

By 45 days postpartum, 952 women (40% of the treatment group) responded to a message asking about their contraceptive use status. Among respondents, 39% reported current use of a modern method. Among non-users, 23% provided a reason for non-use, most often a desire for more information before deciding (43%), exclusive breastfeeding (22%), or concerns about side effects (19%).

At 2.5 months postpartum, 932 women (39%) responded to follow-up messages assessing LAM eligibility. One-third (33%) reported considering the introduction of food to their baby, and among those not introducing food, 41% reported the return of menstruation. These patterns suggest that many women had transitioned out of the effective window for LAM by this stage, underscoring the importance of timely counseling and support for method transition during the postpartum period.

Overall, these figures indicate that overall engagement with the intervention was limited. While nearly all women continued to receive messages and most reported reading them, only a minority actively engaged with the interactive features. Uptake of the shared decision-making counseling flow was particularly low, suggesting that the component most directly designed to support informed contraceptive decision-making reached only a small fraction of participants. This pattern provides important context for interpreting the treatment effects.

3.5 Treatment Effects

3.5.1 Primary Outcomes

Table A8 presents the contraceptive method mix at baseline and follow-up. Short-acting reversible contraceptives (SARCs, primarily injectables and pills) were the most commonly used method type, reported by 29% of women at 3 months postpartum and 30% at 6 months. Use of long-acting and permanent methods (LAPMs) increased from 16% at midline to 23% at endline, indicating a gradual shift toward longer-term options over time. LAM use remained rare at both points, consistent with earlier descriptive evidence on discrepancies between implicit and reported practice.

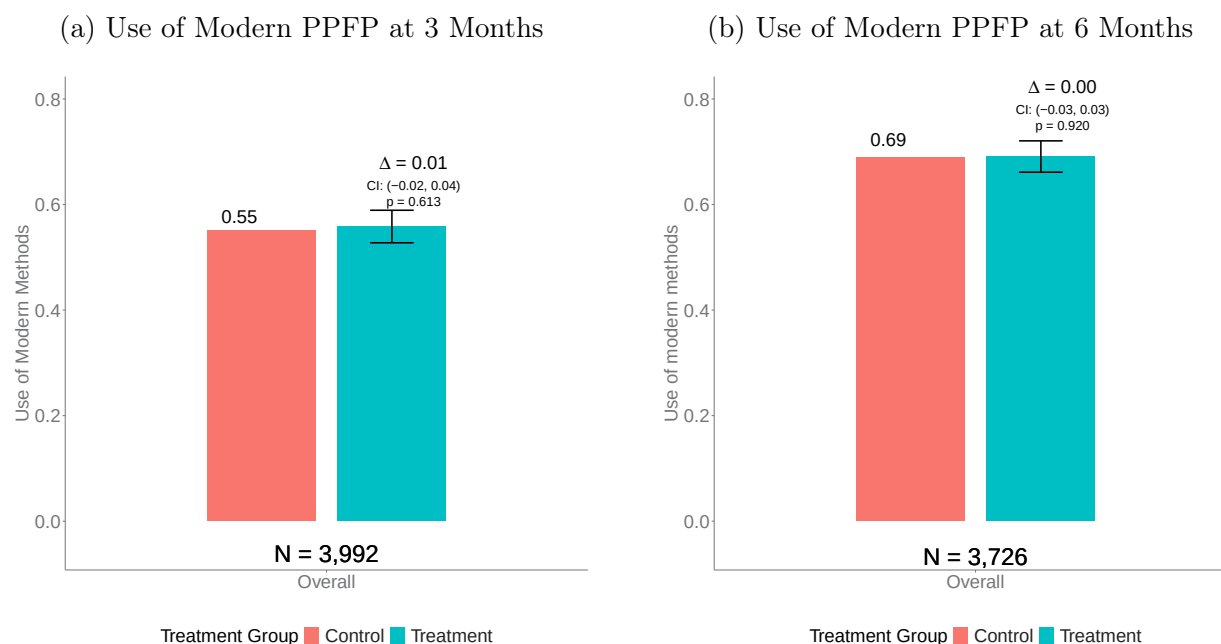
Figure 3 displays the estimated average treatment effects for the two primary outcomes: self-reported use of a modern method at 3 and 6 months postpartum. In the control group, 55% of participants reported modern method use by 3 months, rising to 69% by 6 months postpartum.

The intervention had no detectable effect on modern contraceptive uptake at either time

point. At 3 months postpartum, the estimated treatment effect was 1 percentage point (95% CI: -2 to 4). At 6 months, the estimated effect was 0 percentage points (95% CI: -3 to 3). These null results are precise: we can rule out increases in uptake larger than 7.2 percentage points at midline and 4.3 percentage points at endline, relative to the control group's uptake rates of 55% and 69%.

These findings indicate that enhanced counseling content did not increase contraceptive uptake beyond the already high levels achieved by the standard PROMPTS platform. We explore potential mechanisms behind these null results in Section 3.6.

Figure 3. Primary outcomes



Notes: This figure shows average treatment effects on self-reported postpartum family planning (PPFP) use at 3 months (left) and 6 months (right) postpartum. Bars show the estimated mean outcome in the control group (red bar) and the control group mean plus the estimated treatment effect (blue bar). The treatment effect is represented by the difference between the two bars. Modern methods include sterilization, implants, IUDs, injectables, pills, condoms, diaphragms/cervical caps, spermicides, and the lactational amenorrhea method (LAM). LAM is classified as modern only at 3 months postpartum. Women who experienced pregnancy or infant loss, were interviewed too early (<2.5 months at midline; <5.5 months at endline), reported initiating family planning before their actual delivery date, or had missing or “unsure” responses were excluded from the outcome-specific analyses. Effects were estimated using ordinary least squares with robust standard errors, adjusting for stratification variables (age group and geographic location).

3.5.2 Secondary Outcomes

Figure 4 summarizes the estimated treatment effects for the four prespecified secondary outcomes.

First, the control group scored an average of 0.55 out of 3 on the LAM knowledge index, indicating limited baseline understanding. The intervention produced a statistically signif-

icant improvement of 0.20 points (95% CI: 0.15 to 0.24), corresponding to a 36% increase relative to the control mean. This effect was driven primarily by greater awareness among treatment participants: they were more likely to have heard of LAM (treatment effect: 0.12; 95% CI: 0.09 to 0.15) and to correctly identify that LAM is effective only up to six months postpartum (treatment effect: 0.10; 95% CI: 0.07 to 0.12).

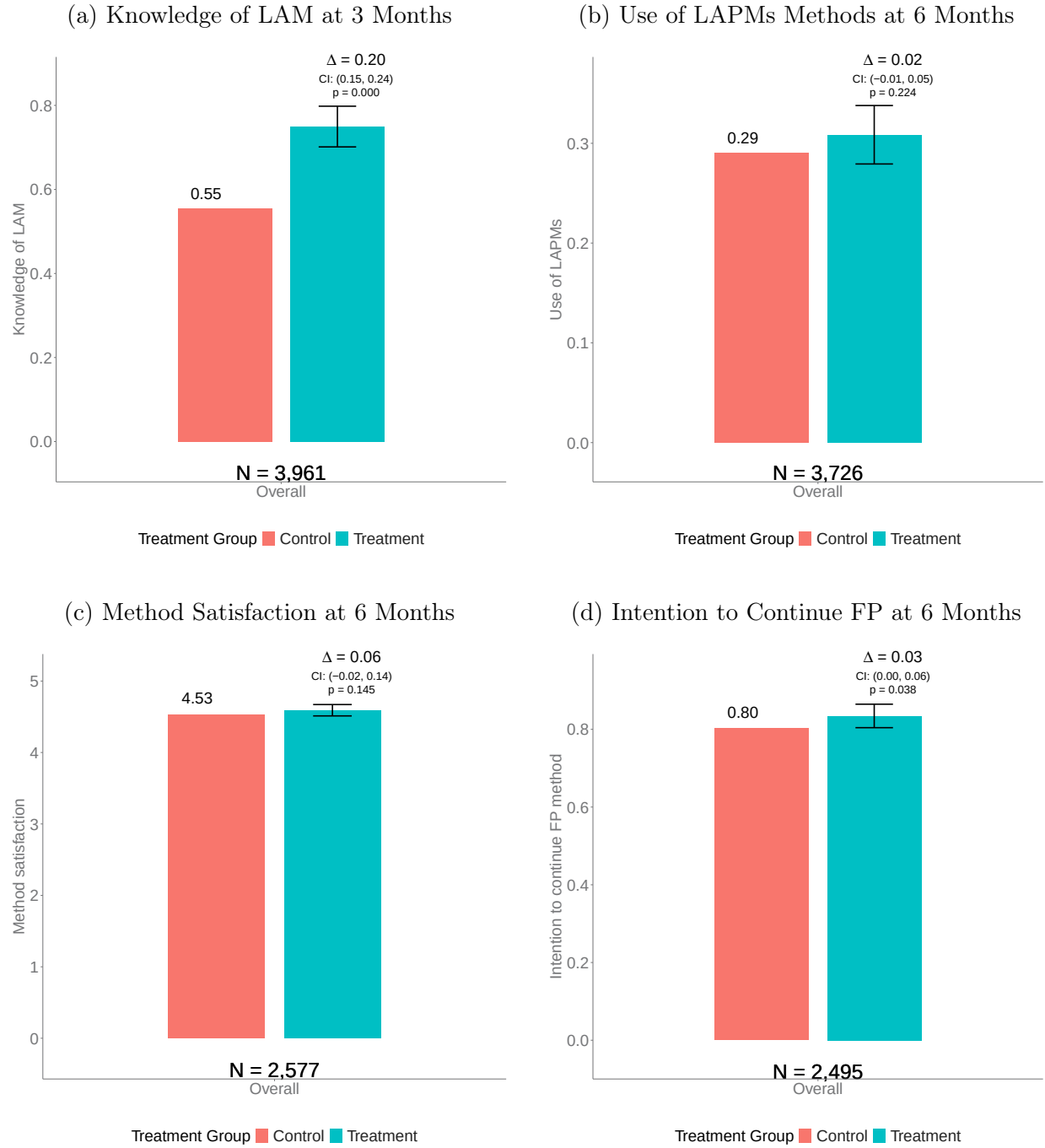
Second, use of LAPMs at six months was 29% in the control group. The estimated treatment effect was a 2 percentage point increase (95% CI: -1 to 5), which is not statistically significant.

Third, method satisfaction at six months was high in the control group (mean: 4.53 out of 5). This aligns with the fact that 87% of family planning users in the control group reported using their preferred method at endline. The intervention had a small, statistically insignificant effect on satisfaction (treatment effect: 0.06 points; 95% CI: -0.02 to 0.14).

Finally, intention to continue family planning at six months was 80% in the control group. The intervention increased this outcome by 3 percentage points (95% CI: 0.0 to 6), representing a 4% improvement relative to the control mean.

These results suggest that the intervention strengthened knowledge of LAM and modestly improved intentions to continue family planning, but had little effect on method satisfaction or uptake of LAPMs.

Figure 4. Secondary outcomes



Notes: This figure presents average treatment effects on four secondary outcomes: knowledge of the lactational amenorrhea method (LAM); use of long-acting or permanent methods (LAPMs)—defined as sterilization, implants, and IUDs; satisfaction with the adopted method; and intention to continue family planning. Bars show the mean outcome in the control group (red bar) and the control group mean plus the estimated treatment effect (blue bar). The difference between the bars represents the treatment effect. LAM knowledge was measured using a three-item index based on knowledge of the three criteria required for effective LAM use. Women were coded as knowing LAM only if they correctly identified all three criteria. “Don’t know” and “unsure” responses were coded as not knowing. Satisfaction and intention outcomes were measured only among women who reported using a family planning method at the time of the survey. Satisfaction was assessed using a Likert scale from 1 (“very unsatisfied”) to 5 (“very satisfied”) and modeled as a continuous variable. Intention was coded as 1 if the respondent indicated that she intended to continue using family planning and 0 if not. LAM knowledge was assessed at 3 months postpartum; the remaining outcomes were measured at 6 months. Women who experienced pregnancy or infant loss, were interviewed too early (before 2.5 months at midline or before 5.5 months at endline), reported initiating family planning before their actual delivery date, or had missing responses were excluded from outcome-specific analyses. Estimates were obtained using ordinary least squares with robust standard errors, adjusting for age group and geographic location (stratification variables). For the LAM knowledge outcome, we additionally controlled for baseline LAM knowledge due to observed imbalances across groups.

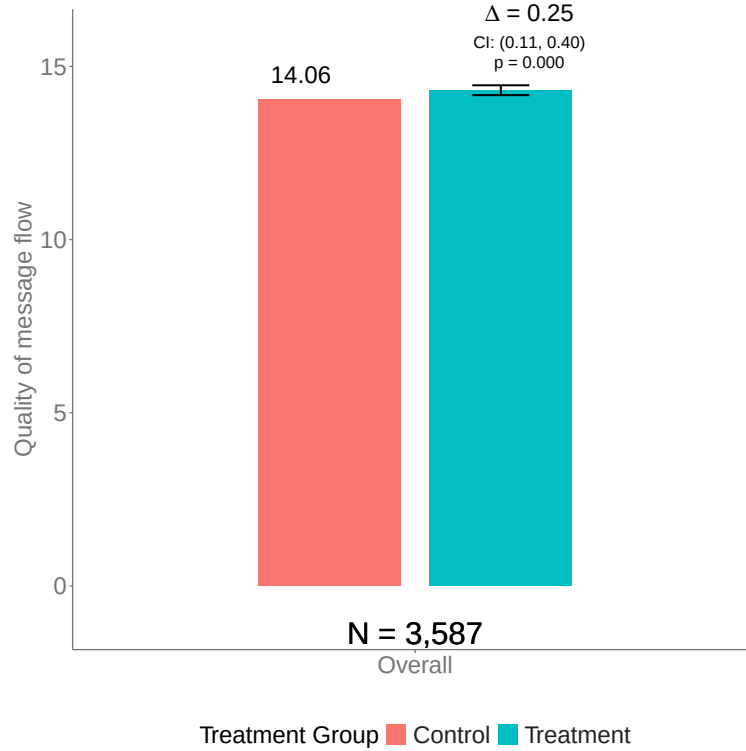
3.5.3 Quality Outcome

Figure 5 reports treatment effects on the overall quality rating of the message flows. Participants rated the messages very highly: in the control group, the mean rating was 14.06 out of 15. Women in the treatment group rated the messages slightly more favorably, with a statistically significant treatment effect of 0.25 points (95% CI: 0.11 to 0.40), representing a 2% improvement relative to the control mean.

Analysis of the three sub-items, shown in Appendix Figures B2a–B2c, suggests that the overall effect was mainly driven by higher ratings in the intervention group on two dimensions: usefulness for learning about contraceptive methods (treatment effect: 0.08; 95% CI: 0.03 to 0.13) and usefulness for choosing a method (treatment effect: 0.12; 95% CI: 0.06 to 0.18). Differences in ratings for the third item, usefulness for addressing family planning concerns, were smaller and not statistically significant (treatment effect: 0.05; 95% CI: –0.00 to 0.11).

Overall, these findings indicate that women in both study arms perceived the message flows as high quality, and that the enhanced counseling content modestly improved perceptions of their usefulness for learning and informed method choice.

Figure 5. User-rated Quality



Notes: This figure shows average treatment effects on user-rated quality of the PROMPTS counseling messages, measured at 6 months postpartum. Bars show the estimated mean outcome in the control group (red bar) and the control group mean plus the estimated treatment effect (blue bar). The treatment effect is represented by the difference between the two bars. The outcome is based on a three-item index (range: 3–15) capturing how helpful the messages were in (1) educating about family planning methods, (2) addressing concerns, and (3) supporting method selection. Each item was rated on a scale from 1 to 5, where 1 indicated “not helpful at all” and 5 indicated “very helpful.” The sample is restricted to women who confirmed receiving PROMPTS messages. Estimates were obtained using ordinary least squares with robust standard errors, controlling for age group and geographic location (stratification variables). Women who were interviewed before 5.5 months postpartum, reported initiating family planning before their actual delivery date, or had missing responses were excluded.

3.6 Mechanisms

We next examine potential explanations for the null average treatment effects on modern contraceptive uptake at 3 and 6 months postpartum. We focus on two areas: patterns of user engagement with the intervention and self-reported reasons for non-use of family planning or preferred methods.

3.6.1 User Engagement

As noted in Section 3.4, engagement with the enhanced counseling content was low and uneven across components. Only a small fraction of treatment group participants completed interactive flows such as the shared decision-making tool. Appendix Table A9 shows, among those who did engage, outcomes were more favorable. At 3 months postpartum, 60% of

SDM users reported modern method use compared to 51% of non-users ($p = 0.004$). At 6 months, uptake of long-acting and permanent methods was 35% among shared decision-making users versus 29% among non-users ($p = 0.029$). Knowledge of LAM was also higher among shared decision-making participants at 3 months (0.84 vs. 0.74; $p = 0.037$). Although these differences are not necessarily causal—given the voluntary and selective nature of engagement—they suggest that the counseling content had value when used. However, the limited reach and uptake of these tools likely dampened the overall impact of the intervention.

Behavioral constraints may have contributed to low engagement. At baseline, 30% of women reported difficulty following through on tasks and 31% reported feeling more strained than usual, patterns that persisted at follow-up. In this context, even low-burden interactive tools may have exceeded available cognitive bandwidth. Nudges and reminders were designed to prompt planning during routine antenatal care, postnatal care, and immunization visits, yet may have gone unnoticed if women were overloaded or preoccupied.

3.6.2 Reasons for Non-Use

We next examine self-reported reasons for not using any contraceptive method, and separately, for not using a preferred method (Table 3). These questions were restricted to women who either did not use any method or were using a method that did not match their stated preference at midline or endline.

Among women not using any method, 72% cited at least one demand-side reason. The most common was fertility preferences (49%), followed by fears of side effects (18%), misinformation or lack of knowledge (13%), clinical ineligibility (8%), planning or time constraints (1%), and social opposition (2.8%). An additional 15% reported that family planning was irrelevant at the time (e.g., not sexually active, awaiting return of menses). Treatment–control differences across these categories were negligible. Notably, only 2.8% of non-users cited social opposition, including male partner disapproval, consistent with high reported partner support (93%).

On the demand side, partner engagement is often highlighted in the literature ([Ashraf et al., 2014](#); [El-Khoury et al., 2016](#); [Link, 2011](#)). In many settings, men play a central role in family planning decision-making, and some interventions suggest that involving male partners can increase uptake. Yet the evidence is mixed: for example, [Harrington et al. \(2019\)](#) found no significant effect of optional male involvement in the Mobile WACH XY intervention, and [Karra et al. \(2021\)](#) showed that partner presence during counseling did not improve concordance between women’s preferences and contraceptive behavior. Our findings are consistent with this latter body of work: fewer than 3% of women cited social opposition as a reason for non-use, suggesting that male partner opposition is not a major barrier in

this context. Instead, fertility preferences, misinformation, and side-effect concerns remain central. The intervention was designed to address misinformation and side-effect concerns, but limited engagement likely constrained its effectiveness. Fertility preferences, by contrast, were not a target of the intervention and therefore represent a structural source of non-use beyond its scope.

Supply-side reasons were less common overall (6.0%), but slightly more frequent in the treatment group (7.1% vs. 4.9%, $p = 0.058$), largely driven by reports of method availability or access (4.8% vs. 2.9%, $p = 0.048$). These results align with prior work suggesting that supply-side and financial barriers are often overstated as explanations for low contraceptive uptake ([Dupas et al., 2025](#); [Miller et al., 2025](#)).

3.6.3 Preferred Method Mismatch

A core aim of the intervention was to improve concordance between contraceptive use and women’s stated preferences. Most women in the study did use their preferred method, but among those who did not, supply-side barriers were dominant. Sixty-seven percent cited at least one supply-side reason, most often provider unavailability or denial (36%) and method access issues (31%). Demand-side explanations were less common, with 24% citing clinical ineligibility and fewer than 6% citing fears or social opposition. These findings are consistent with provider surveys in Kenya and comparable contexts, which document provider bias in restricting certain methods based on age, parity, or marital status ([Wagner et al., 2025](#); [Schwandt et al., 2017](#); [Tumlinson et al., 2015](#)). Again, treatment–control differences were negligible.

Taken together, these findings suggest that the null effects on contraceptive uptake stem from a combination of limited engagement, intrinsic preferences against use, and supply-side barriers beyond the intervention’s scope. At the same time, improvements in knowledge, intentions, and perceived quality highlight the potential for digital counseling tools to support PPFP when engagement is higher or when structural barriers are less binding.

Table 3. Reason for lack of uptake of any method or a preferred method

Characteristic	Overall N = 2,226	Control N = 1,135	Treatment N = 1,091	p-value
Demand-side reasons for lack of contraceptive use				
Any demand-side reason	1,206 / 1,668 (72%)	614 / 854 (72%)	592 / 814 (73%)	0.7
Social opposition	46 / 1,668 (2.8%)	23 / 854 (2.7%)	23 / 814 (2.8%)	0.9
Knowledge/misperceptions	217 / 1,668 (13%)	105 / 854 (12%)	112 / 814 (14%)	0.4
Fears of side effects/infertility/breastfeeding	296 / 1,668 (18%)	146 / 854 (17%)	150 / 814 (18%)	0.5
Preferences & fertility intentions	824 / 1,668 (49%)	428 / 854 (50%)	396 / 814 (49%)	0.5
Planning/time constraints	20 / 1,668 (1.2%)	10 / 854 (1.2%)	10 / 814 (1.2%)	>0.9
Clinical/health ineligibility	131 / 1,668 (7.9%)	67 / 854 (7.8%)	64 / 814 (7.9%)	>0.9
FP irrelevant	255 / 1,668 (15%)	143 / 854 (17%)	112 / 814 (14%)	0.090
Supply-side reasons for lack of contraceptive use				
Any supply-side reason	100 / 1,668 (6.0%)	42 / 854 (4.9%)	58 / 814 (7.1%)	0.058
Supply: method access/availability	64 / 1,668 (3.8%)	25 / 854 (2.9%)	39 / 814 (4.8%)	0.048
Supply: provider availability/denial	9 / 1,668 (0.5%)	3 / 854 (0.4%)	6 / 814 (0.7%)	0.3
Supply: financial cost/affordability	29 / 1,668 (1.7%)	14 / 854 (1.6%)	15 / 814 (1.8%)	0.8
Demand-side reasons for not using preferred method				
Any demand-side reason	207 / 641 (32%)	105 / 323 (33%)	102 / 318 (32%)	>0.9
Social opposition	22 / 641 (3.4%)	14 / 323 (4.3%)	8 / 318 (2.5%)	0.2
Clinical/health ineligibility	156 / 641 (24%)	78 / 323 (24%)	78 / 318 (25%)	>0.9
Fears of side effects/infertility	34 / 641 (5.3%)	18 / 323 (5.6%)	16 / 318 (5.0%)	0.8
Supply-side reasons for not using preferred method				
Any supply-side reason	432 / 641 (67%)	220 / 323 (68%)	212 / 318 (67%)	0.7
Supply: method access/availability	199 / 641 (31%)	96 / 323 (30%)	103 / 318 (32%)	0.5
Supply: provider availability/denial	232 / 641 (36%)	122 / 323 (38%)	110 / 318 (35%)	0.4
Supply: financial cost/affordability	29 / 641 (4.5%)	16 / 323 (5.0%)	13 / 318 (4.1%)	0.6

Notes: This table reports reasons for not using any contraceptive method and, separately, for not using a preferred method. Denominators differ across panels. For the lack of contraceptive use rows, the sample is restricted to women who reported not using any method at either midline or endline ($N = 1,668$). For the not using preferred method rows, the sample is restricted to women who reported using a method but not their preferred method ($N = 641$). Within each panel, respondents could select multiple reasons, so percentages do not sum to 100.

Demand-side reasons for lack of contraceptive use include: (i) social opposition (disapproval from a partner, family, or religious authorities); (ii) knowledge/misperceptions (doubts about effectiveness, not knowing how to use methods, or being undecided about which method to use); (iii) fears (concerns about side effects, infertility, or breastfeeding); (iv) preferences and fertility intentions (not wanting to use contraception or wanting another child); (v) planning/time constraints (lack of time to seek services); (vi) clinical/health ineligibility (medical conditions or postpartum recovery); (vii) family planning irrelevant (not sexually active, living apart from a partner, or awaiting return of menses).

Supply-side reasons for lack of contraceptive use include: (i) method access/availability (stock-outs or unavailable methods); (ii) provider availability/denial (provider absent or refusing services); (iii) financial cost/affordability (methods considered too expensive).

For the not using preferred method rows, demand-side categories mirror those above, except that fears and clinical/health ineligibility reflect concerns specific to the preferred method. Supply-side reasons also mirror those above but refer specifically to barriers to obtaining the preferred method (such as stock-outs, provider refusal, or cost).

Reported p -values compare proportions between treatment and control arms using two-sided tests.

3.7 Treatment Effect Heterogeneity

While the average treatment effects on postpartum contraceptive uptake were small and statistically insignificant, these results may conceal important heterogeneity across subgroups. We therefore estimate treatment effects within subpopulations that, based on prior literature and theoretical expectations, would be most likely to respond to the intervention.

3.7.1 Geographic Differences

We first examine heterogeneity by region. Women in Central Kenya generally have higher socioeconomic status and educational attainment, whereas women in Coastal Kenya face multiple disadvantages, including higher poverty rates and weaker health systems. As shown in Figure 6, we find no differential treatment effects across regions. The estimates are imprecise, however: at three months postpartum we cannot rule out effects as large as a 8 percentage point increase in Western Kenya, and at six months we cannot rule out effects ranging from a 13 percentage point decrease in Coastal Kenya.

3.7.2 Fertility Preferences and Intentions

We then analyze heterogeneity by fertility preferences. One set of subgroups includes women who, at baseline, reported not wanting more children. These women face the highest costs of another unintended pregnancy and might therefore be more motivated to adopt family planning. We also examine women who reported an intention to start using a contraceptive method within six months postpartum. As shown in Figures 7 and 8, we find no evidence of differential treatment effects in these groups.

3.7.3 Financial Constraints

Next, we stratify by a proxy for baseline purchasing power. Specifically, women were asked whether their household could easily raise 2,000 KSh (approximately 20 USD) for medical treatment if needed. As shown in Figure 9, we again find no meaningful differences in treatment effects across wealth levels, consistent with prior studies showing financial constraints were not the primary barrier in this setting (Dupas et al., 2025).

3.7.4 Access to Care

We also investigate whether the effects varied by proximity to health facilities. We use an indicator for whether the respondent walked to her antenatal care clinic as a proxy for distance. Figure 10 shows no evidence of differential effects by distance. The estimates are imprecise, however, and we cannot rule out treatment effects as large as 10 percentage points (18% relative to the control mean).

3.7.5 Behavioral Constraints

Finally, we examine whether the intervention was more effective among women who may face behavioral barriers to follow-through. Since most women in the sample expressed an

intention to adopt a method postpartum, lack of follow-through may stem from procrastination, forgetfulness, or competing demands. As shown in Figure 11, we find no evidence of stronger effects among these women.

Across all subgroups, we fail to reject the null hypothesis of no treatment effect. Although limited sample sizes in some strata reduce precision, the confidence intervals in most cases rule out large positive or negative effects. These analyses reinforce the main finding that the intervention did not increase modern postpartum contraceptive uptake, even among groups that theory and prior evidence suggest would be most likely to benefit.

Figure 6. Treatment Effects on Modern Contraceptive Uptake by Geographic Region

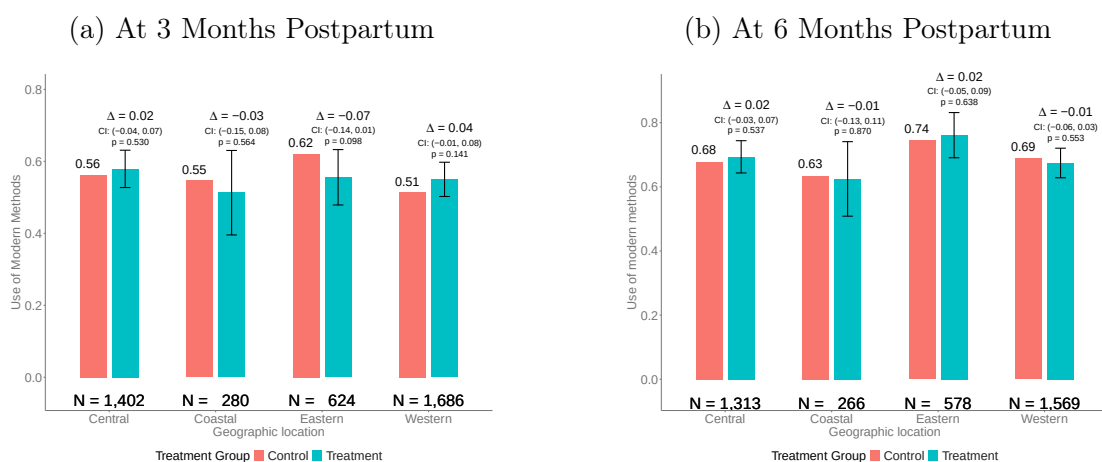


Figure 7. Treatment Effects on Modern Contraceptive Uptake by Desire for Another Child

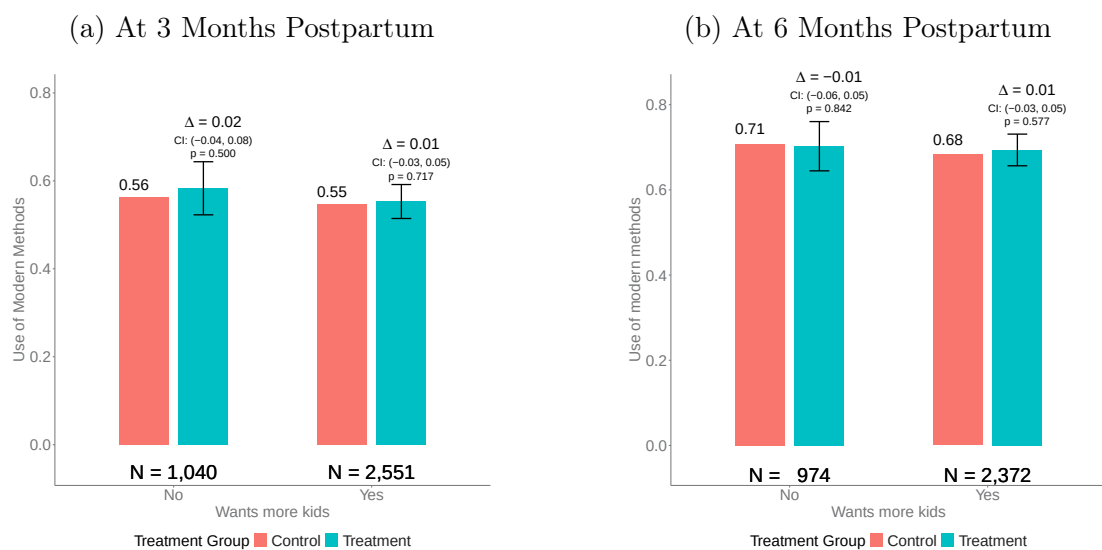


Figure 8. Treatment Effects on Modern Contraceptive Uptake by Baseline Intention to Adopt Within Six Months

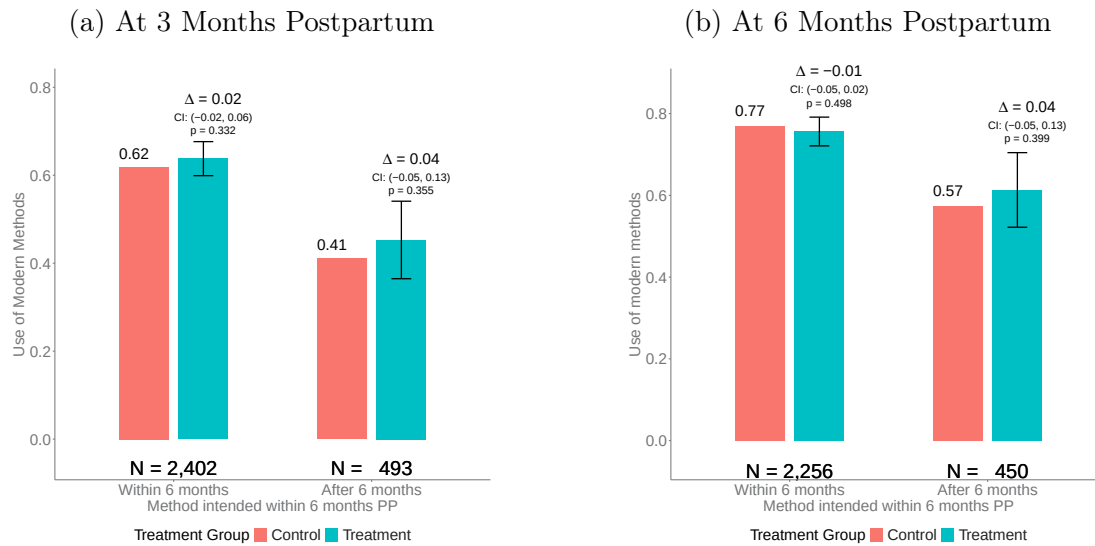


Figure 9. Treatment Effects on Modern Contraceptive Uptake by Household Ability to Raise 2,000 KSh

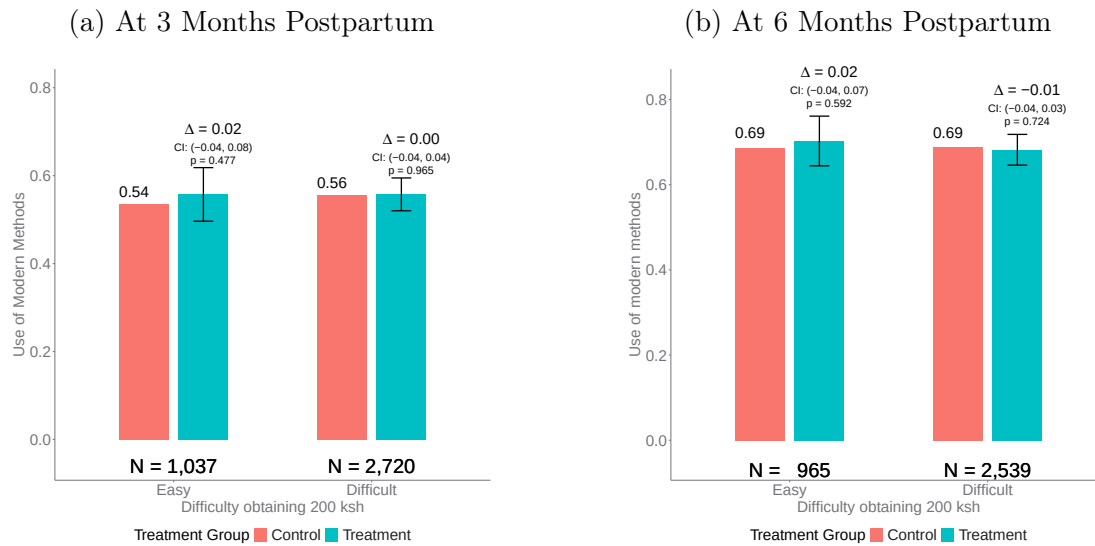


Figure 10. Treatment Effects on Modern Contraceptive Uptake by Distance to Antenatal Care (ANC) Facility

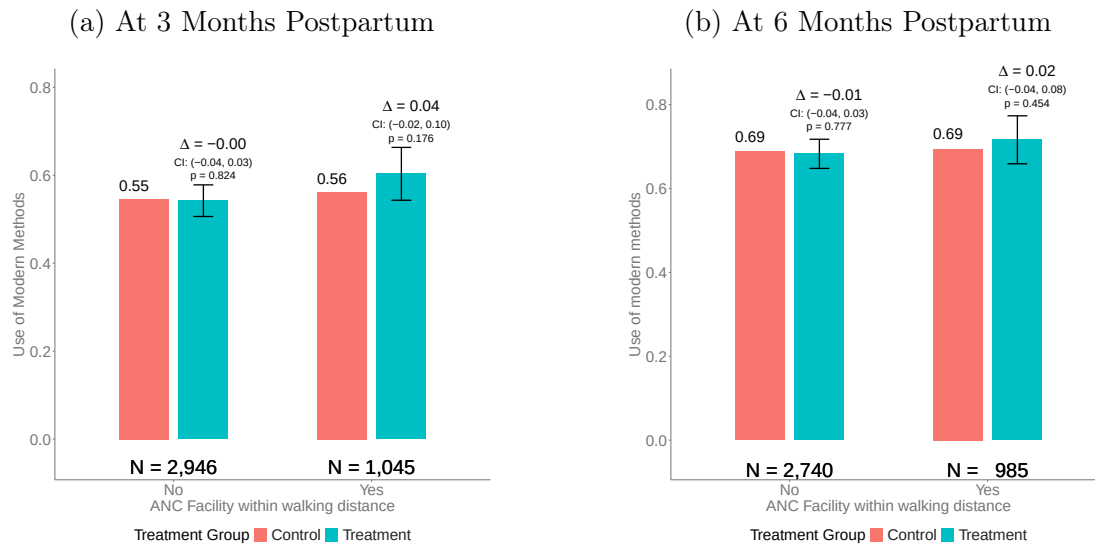
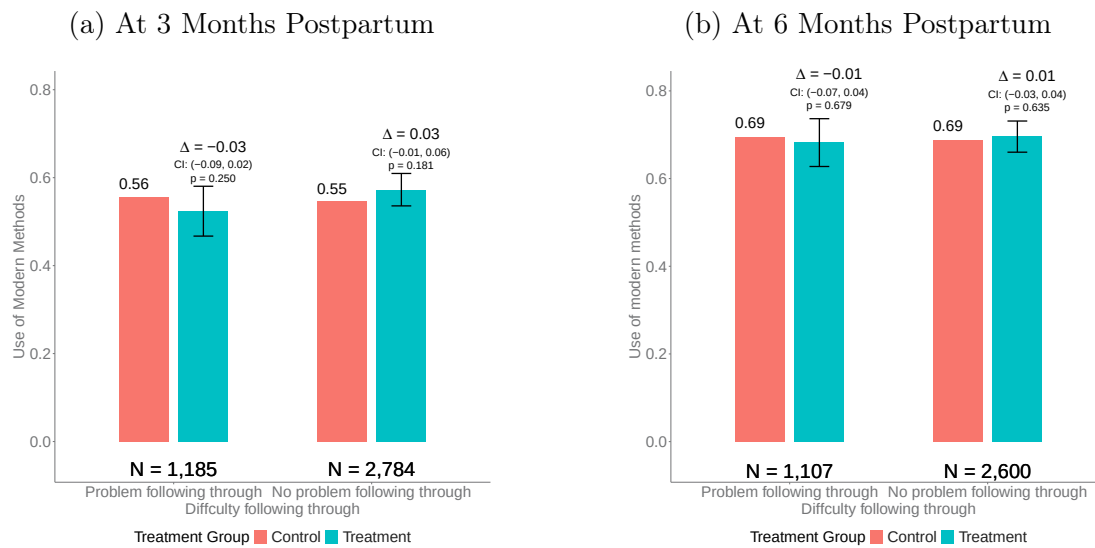


Figure 11. Treatment Effects on Modern Contraceptive Uptake by Reported Difficulty Following Through



4 Discussion

This study evaluated the impact of a digital, personalized counseling intervention delivered via SMS text messages on postpartum family planning in Kenya. The intervention improved several secondary outcomes—including knowledge of the lactational amenorrhea method (LAM), intention to continue family planning, and perceptions of counseling quality—but did not increase modern contraceptive use at three or six months postpartum, the trial’s primary outcomes.

Analysis of mechanisms highlights two main explanations for the null results. First, user engagement with the enhanced counseling content was limited: only 20% of women interacted with the shared decision-making tool, and 6% completed the full counseling flow. This low uptake is consistent with prior evidence that digital health interventions often face participation challenges due to message fatigue, cognitive overload, and competing demands (LeFevre et al., 2017; Aung et al., 2020). Second, among women not using any method, nearly half reported no desire to avoid pregnancy, indicating that fertility preferences were an important determinant of non-use. Other demand-side barriers, including fears of side effects, misinformation, and concerns related to breastfeeding, were also common. Although these issues were targeted by the intervention, limited interaction likely constrained effectiveness. Supply-side barriers were reported slightly more often by treatment participants but remained infrequent overall.

Among the subset of women who engaged with the shared decision-making tool, contraceptive uptake and knowledge outcomes were significantly better, suggesting that the content was effective when used. This pattern indicates that low engagement, rather than lack of content relevance, limited the intervention’s impact.

These findings align with previous studies that show limited or no effect of digital counseling tools on contraceptive uptake, despite improvements in knowledge and user experience (Dehlendorf et al., 2019; Johnson et al., 2017). For example, Athey et al. (2023) report large effects on long-acting reversible contraceptive (LARC) uptake from a provider-led shared decision-making intervention in Cameroon. Unlike that study, which was facility-based and delivered by trained providers, the current trial tested a fully remote, text-based intervention at national scale. In addition, the intervention emphasized method choice aligned with women’s stated preferences rather than promoting specific methods such as LARCs. Exploratory analyses suggested positive effects on LARC uptake in Central Kenya, but such increases are meaningful only insofar as they reflect informed, voluntary choices.

Although there was no effect on contraceptive use, the intervention’s secondary impacts remain noteworthy. Knowledge of LAM and intentions to continue family planning may

be more responsive to light-touch exposure and less dependent on sustained interaction. Clarifying LAM criteria through brief messages may have improved understanding, and personalized framing may have enhanced women’s sense of support and trust. Such changes could shape longer-term contraceptive behavior, particularly if reinforced through provider contact or subsequent exposure to family planning services.

The intervention was also inexpensive. The marginal cost of the enhanced counseling content relative to the base PROMPTS package was approximately \$0.74 per participant, indicating potential for cost-effective delivery if paired with strategies to increase engagement.

Several limitations merit consideration. First, the timing of messages was anchored to estimated delivery dates collected at enrollment. Although actual delivery dates were later incorporated into survey timing, some messages may have been received too early or too late to be salient. Second, outcomes were self-reported and may be subject to recall bias or social desirability, particularly for sensitive measures such as contraceptive use. Third, participants were women who opted into the PROMPTS platform and therefore differed from a nationally representative cohort. Compared with women in national survey data, participants were more educated and more likely to have prior family planning experience, suggesting higher baseline awareness and narrower informational gaps. In more disadvantaged populations, effects may differ depending on the relative importance of knowledge versus access barriers. Fourth, all participants received the standard PROMPTS package, which has previously been shown to increase postpartum contraceptive use ([Jones et al., 2020](#)). Uptake in the control group reached 69% at six months postpartum, substantially higher than the national average of 50% ([Track20 Project, 2021](#)), leaving limited scope for further improvements. Finally, follow-up ended at 6.5 months postpartum. Although this captures the period when contraceptive initiation is most clinically relevant, it does not allow assessment of longer-term outcomes, which may be influenced by the observed improvements in knowledge, intentions, and perceptions of counseling quality.

5 Conclusion

This study finds that a remote, personalized digital counseling intervention delivered via text messages improved knowledge of the lactational amenorrhea method, increased intention to use family planning, and enhanced perceived quality of counseling, but did not increase postpartum contraceptive uptake at the population level. These findings underscore both the promise and the limitations of digital counseling for postpartum family planning.

A central takeaway is that making a remote digital counseling tool available is not suffi-

cient to shift behavior at scale. Limited engagement, particularly during periods of low cognitive bandwidth such as pregnancy and the postpartum period, appears to be a key barrier. Future research should examine how specific features of message flows, including content and timing, can be optimized to maximize user engagement. In parallel, there is a need to develop models that connect remote counseling with in-person support from healthcare providers. This is especially important given evidence from this study that supply-side constraints are a primary reason why some contraceptive users are unable to obtain their preferred method. [Holt et al. \(2020\)](#) found that a decision support tool encouraged providers to incorporate clients' preferences and values into counseling discussions, rather than emphasizing specific methods in isolation. Integrating digital tools with provider-led counseling may increase the relevance and effectiveness of family planning services, particularly in facility-based care.

Future studies should also assess the persistence of effects over time, as contraceptive needs evolve beyond the early postpartum period. It is important to recognize, however, that a key finding from this study and from recent literature is that high desired fertility remains a major reason for non-use of contraception ([Dupas et al., 2025](#)). Digital interventions can help reduce informational, behavioral, or access-related barriers, but they are unlikely to alter fertility preferences, especially in contexts where contraceptive uptake is already high at baseline. Thus, even with improvements in design and delivery, the scope of digital counseling tools may remain limited to addressing second-order barriers such as misinformation, uncertainty, or provider interaction quality, rather than structural or deeply held demand-side factors like fertility preferences.

A Appendix A: Tables

Appendix Table A1. Baseline Summary Statistics - Midline Sample

Characteristic	Overall N = 4,167	Control N = 2,088	Treatment N = 2,079	p-value
Demographic and Socioeconomic Characteristics				
Age (years)	26.26 (5.72)	26.30 (5.76)	26.22 (5.67)	0.7
Aged 15–19 (adolescent)	8.5%	8.0%	8.9%	0.3
Married	79%	79%	80%	0.7
Cohabiting, unmarried	1.2%	1.0%	1.4%	0.3
Highest education: Tertiary	31%	31%	30%	0.8
Highest education: Secondary	49%	49%	50%	0.5
Highest education: Primary	19%	20%	19%	0.5
Highest education: None	0.8%	0.8%	0.9%	0.7
Location: Western Kenya	42%	42%	43%	0.5
Location: Central Kenya	35%	36%	35%	0.6
Location: Eastern Kenya	16%	16%	15%	0.7
Location: Coastal Kenya	6.9%	6.8%	7.1%	0.7
Ever given birth (live or stillbirth)	58%	57%	58%	0.5
Number of living children	1.78 (1.06)	1.81 (1.09)	1.75 (1.03)	0.2
FP Knowledge and Beliefs				
Aware of return to fertility postpartum	36%	35%	37%	0.13
Knowledge index: LAM	0.51 (0.76)	0.49 (0.75)	0.54 (0.77)	0.037
Husband supportive of PPFP	93%	93%	93%	0.8
Behavioral and Psychological Factors				
Needs reminders to start tasks	13%	13%	13%	0.9
Difficulty planning ahead	20%	19%	21%	0.3
Difficulty following through	30%	29%	31%	0.2
Felt more strained than usual (past month)	31%	31%	30%	0.6
Access and Financial Constraints				
Travel time to ANC facility (min)	29.16 (22.67)	28.79 (22.48)	29.54 (22.86)	0.3
Obtaining 2,000 KSh for care: difficult	0.72 (0.45)	0.73 (0.44)	0.72 (0.45)	0.4
FP History and Intentions				
Current pregnancy was unplanned	34%	34%	34%	0.7
Does not want more children	29%	28%	29%	0.5
Wants to wait 2+ years before next pregnancy	97%	96%	97%	0.4
Prior FP use	64%	64%	64%	0.8
Last method used was modern	61%	60%	61%	0.7
Last FP source: public facility	34%	33%	35%	0.2
Last FP source: private facility	11%	11%	12%	0.5
Last FP source: pharmacy	19%	19%	18%	0.2
Intends to use PPFP	85%	85%	85%	0.6
Plans to use a modern method postpartum	70%	68%	71%	0.012
Plans to start method within 6 months postpartum	69%	69%	68%	0.8
Weeks to EDD	7.59 (3.31)	7.60 (3.29)	7.59 (3.34)	0.8
Test of joint orthogonality, p-value				0.2

Abbreviations: PMA = Performance Monitoring for Action; FP = family planning; LAM = lactational amenorrhea method; PPFP = postpartum family planning; ANC = antenatal care; KSh = Kenyan shillings; EDD = expected delivery date.

Notes: This table presents baseline characteristics of participants, overall and stratified by treatment assignment. Means and standard deviations (SD) are shown for continuous variables; frequencies and percentages for categorical variables. P-values are from chi-square tests for categorical variables and Wilcoxon rank-sum tests for continuous variables, testing for differences between treatment and control groups. The joint F test p -value, based on randomization inference, is reported as an overall assessment of baseline covariate balance. Missing values are excluded from variable-specific denominators. Randomization was stratified by age group and geographic location.

Appendix Table A2. Baseline Summary Statistics - Endline Sample

Characteristic	Overall N = 3,743	Control N = 1,884	Treatment N = 1,859	p-value
Demographic and Socioeconomic Characteristics				
Age (years)	26.34 (5.69)	26.38 (5.73)	26.30 (5.65)	0.7
Aged 15–19 (adolescent)	7.9%	7.4%	8.4%	0.3
Married	80%	80%	80%	0.7
Cohabiting, unmarried	1.2%	1.1%	1.4%	0.4
Highest education: Tertiary	31%	32%	31%	0.5
Highest education: Secondary	49%	48%	50%	0.2
Highest education: Primary	19%	20%	19%	0.4
Highest education: None	0.8%	0.8%	0.8%	0.9
Location: Western Kenya	42%	42%	42%	0.8
Location: Central Kenya	35%	35%	35%	>0.9
Location: Eastern Kenya	16%	16%	15%	0.6
Location: Coastal Kenya	7.1%	7.1%	7.2%	>0.9
Ever given birth (live or stillbirth)	58%	58%	59%	0.7
Number of living children	1.78 (1.06)	1.81 (1.09)	1.75 (1.04)	0.2
FP Knowledge and Beliefs				
Aware of return to fertility postpartum	36%	35%	38%	0.2
Knowledge index: LAM	0.52 (0.76)	0.49 (0.75)	0.54 (0.78)	0.027
Husband supportive of PPFP	93%	94%	93%	0.8
Behavioral and Psychological Factors				
Needs reminders to start tasks	13%	13%	13%	0.9
Difficulty planning ahead	20%	19%	21%	0.3
Difficulty following through	30%	29%	31%	0.12
Felt more strained than usual (past month)	30%	31%	30%	0.8
Access and Financial Constraints				
Travel time to ANC facility (min)	28.92 (22.44)	28.44 (22.36)	29.41 (22.52)	0.13
Obtaining 2,000 KSh for care: difficult	0.72 (0.45)	0.73 (0.44)	0.72 (0.45)	0.5
FP History and Intentions				
Current pregnancy was unplanned	34%	33%	34%	0.4
Does not want more children	29%	28%	30%	0.3
Wants to wait 2+ years before next pregnancy	97%	96%	97%	0.3
Prior FP use	65%	64%	65%	0.8
Last method used was modern	61%	61%	62%	0.8
Last FP source: public facility	35%	34%	35%	0.3
Last FP source: private facility	11%	11%	12%	0.7
Last FP source: pharmacy	19%	19%	18%	0.4
Intends to use PPFP	85%	85%	85%	0.5
Plans to use a modern method postpartum	70%	68%	72%	0.006
Plans to start method within 6 months postpartum	69%	69%	69%	>0.9
Weeks to EDD	7.61 (3.31)	7.62 (3.28)	7.59 (3.34)	0.6
Test of joint orthogonality, p-value				0.3

Abbreviations: PMA = Performance Monitoring for Action; FP = family planning; LAM = lactational amenorrhea method; PPFP = postpartum family planning; ANC = antenatal care; KSh = Kenyan shillings; EDD = expected delivery date.

Notes: This table presents baseline characteristics of participants, overall and stratified by treatment assignment. Means and standard deviations (SD) are shown for continuous variables; frequencies and percentages for categorical variables. P-values are from chi-square tests for categorical variables and Wilcoxon rank-sum tests for continuous variables, testing for differences between treatment and control groups. The joint *F* test *p*-value, based on randomization inference, is reported as an overall assessment of baseline covariate balance. Missing values are excluded from variable-specific denominators. Randomization was stratified by age group and geographic location.

Appendix Table A3. Descriptive Statistics at Midline

Characteristic	Overall N = 4,167	Control N = 2,088	Treatment N = 2,079
Pregnancy and Postpartum Outcomes			
Delivered, baby alive	96%	96%	96%
Delivered alive, but baby no longer alive	2.5%	2.3%	2.6%
Delivered a stillbirth	0.7%	0.7%	0.8%
Had a miscarriage	0.6%	0.6%	0.6%
Breastfeeding and Fertility Indicators			
Currently breastfeeding	100%	100%	100%
No more than 4 hours between breastfeeds	96%	96%	96%
No more than 6 hours between breastfeeds	99%	99%	99%
Baby given other food or liquids	17%	16%	18%
Menstrual periods have returned	53%	55%	51%
Meets all three LAM criteria	36%	35%	36%
FP Counseling and Health Contacts			
Discussed FP during ANC	59%	59%	58%
Discussed FP in last ANC month	49%	48%	49%
Discussed FP during delivery care	55%	55%	55%
Number of postpartum checkups	1.29 (1.30)	1.31 (1.34)	1.27 (1.26)
Weeks until first postpartum checkup	3.94 (2.48)	3.89 (2.56)	4.00 (2.40)
Discussed FP at postpartum visit	71%	71%	70%
Discussed FP at child immunization visit	65%	65%	64%
Access and Behavioral Factors			
Experienced difficulty accessing FP	3.4%	3.4%	3.4%
Needs reminders to start tasks	7.5%	7.4%	7.7%
Difficulty planning ahead	12%	12%	12%
Difficulty following through	22%	23%	22%
Felt more strained than usual in past month	25%	24%	25%

Abbreviations: FP = family planning; LAM = lactational amenorrhea method; ANC = antenatal care; PP = postpartum.

Notes: This table summarizes selected participant characteristics reported at the 3-month follow-up (midline). Means and standard deviations (SD) are shown for continuous variables; frequencies and percentages are reported for categorical variables. Missing values are excluded from variable-specific denominators.

The variable “Implicit LAM use” captures women who meet the clinical criteria for LAM effectiveness: (i) the baby is under 6 months old, (ii) the woman is exclusively breastfeeding (no food or other fluids introduced), and (iii) menstruation has not resumed. This variable is constructed based on reported behaviors, not self-identification.

Appendix Table A4. Descriptive Statistics at Endline

Characteristic	Overall N = 3,743	Control N = 1,884	Treatment N = 1,859
FP Counseling and Health Contacts			
Number of postpartum visits since midline	1.02 (1.19)	1.02 (1.21)	1.02 (1.18)
Discussed FP at a postpartum visit	77%	77%	77%
Discussed FP at child immunization visit	56%	57%	55%
Behavioral and Psychological Factors			
Needs reminders to start tasks	8.6%	8.4%	8.7%
Difficulty planning ahead	13%	13%	13%
Difficulty following through	23%	22%	24%
Felt more strained than usual in past month	23%	23%	23%
Exposure to and Engagement with PROMPTS Messages			
Currently receiving PROMPTS messages	98%	98%	98%
Always reads PROMPTS messages	81%	81%	81%
Often reads PROMPTS messages	14%	14%	15%
Partner is enrolled in PROMPTS	19%	18%	19%
Partner is receiving PROMPTS messages	82%	83%	81%

Abbreviations: FP = family planning.

Notes: This table summarizes selected participant characteristics reported at the 6-month follow-up (endline). Means and standard deviations (SD) are shown for continuous variables; frequencies and percentages are reported for categorical variables. Missing values are excluded from variable-specific denominators.

Appendix Table A5. Treatment Group Engagement with the Initial Nudge Message Flows

Characteristic	N = 2,376
Any user engagement	463 / 2,374 (20%)
Exclusive breastfeeding	81 / 463 (17%)
Implant contraception	76 / 463 (16%)
Depo-Provera injection	80 / 463 (17%)
Traditional birth control pill	47 / 463 (10%)
Condom use	22 / 463 (4.8%)
Copper IUD	160 / 463 (35%)
Rhythm beads	51 / 463 (11%)
Mini pill	30 / 463 (6.5%)
Hormonal IUD	50 / 463 (11%)

Abbreviations: IUD = intrauterine device.

Notes: This table summarizes engagement with the initial counseling messages sent at approximately 7.5 months of pregnancy to participants in the treatment group (N = 2,376). These messages introduced postpartum family planning and provided brief educational content on multiple contraceptive methods. Women were invited to respond if they were interested in learning more. Engagement is defined as any texted response to the initial message flow. Among the 463 women (20%) who responded, follow-up messages provided more detailed information tailored to their interest. The most commonly selected methods were copper IUD (35%), exclusive breastfeeding (17%), and the Depo-Provera injection (17%). In each row, the numerator/denominator and percentage indicate the number and share of women who responded to that specific prompt, not the full treatment-group denominator.

Appendix Table A6. Treatment Group Engagement with the SDM Message Flows

Characteristic	N = 2,376
User selected preferred method attribute(s)	423 / 2,123 (20%)
Prefer method that is highly effective	114 / 423 (27%)
Prefer method that lasts a long time	90 / 423 (21%)
Prefer method that allows regular periods	68 / 423 (16%)
Prefer method safe while breastfeeding	152 / 423 (36%)
Prefer method stoppable for future pregnancy	128 / 423 (30%)
Suggested methods offered	322 / 2,123 (15%)
Suggested method: Exclusive breastfeeding (LAM)	32 / 322 (9.9%)
Suggested method: Implant	87 / 322 (27%)
Suggested method: Depo (Injection)	71 / 322 (22%)
Suggested method: Traditional pill	15 / 322 (4.7%)
Suggested method: Condoms	8 / 322 (2.5%)
Suggested method: Copper IUD/coil	27 / 322 (8.4%)
Suggested method: Rhythm/Cycle beads	11 / 322 (3.4%)
Suggested method: Mini Pill	55 / 322 (17%)
Suggested method: Hormonal IUD	10 / 322 (3.1%)
Suggested method: IUD (Copper or Hormonal)	6 / 322 (1.9%)
Method most interested in discussing with provider	
Implant	50 / 135 (37%)
Depo (Injection)	26 / 135 (19%)
Mini pill	19 / 135 (14%)
Copper IUD/coil	11 / 135 (8.1%)
Exclusive breastfeeding (Lactational Amenorrhea)	9 / 135 (6.7%)
Rhythm method/Cycle beads	7 / 135 (5.2%)
Traditional birth control pill	6 / 135 (4.4%)
IUD (Copper or Hormonal)	4 / 135 (3.0%)
Condoms	2 / 135 (1.5%)
Hormonal IUD	1 / 135 (0.7%)

Abbreviations: SDM = shared decision-making; LAM = lactational amenorrhea method; IUD = intrauterine device; Depo = Depo-Provera injectable contraceptive.

Notes: This table summarizes engagement with the SDM message flows initiated at approximately 8 months of gestation among women in the treatment group (N = 2,381). Women were asked to identify their most valued method attributes, and the platform recommended up to three methods using a decision tree. The final step of the SDM flow asked users to select the method they were most interested in discussing with a provider. Engagement levels reflect actual message-interaction data recorded through the PROMPTS system. In each row, the numerator/denominator and percentage indicate the number and share of women who responded to that specific message prompt (i.e., provided a response), not the entire treatment-group denominator.

Appendix Table A7. Treatment Group Engagement with Postpartum Message Flows

Characteristic	N = 2,376
21 days postpartum	
Selected topics to learn about	226 / 2,107 (11%)
Topics selected to learn – Timing	43 / 226 (19%)
Topics selected to learn – Safety	31 / 226 (14%)
Topics selected to learn – Traditional methods	22 / 226 (9.7%)
Topics selected to learn – Resuming sex	51 / 226 (23%)
Topics selected to learn – Family planning methods	81 / 226 (36%)
36 days postpartum	
Confirm still interested in chosen method	34 / 105 (32%)
Method selected to learn more about	111 / 1,716 (6.5%)
Selected to learn: Exclusive breastfeeding (LAM)	11 / 111 (9.9%)
Selected to learn: Implant	22 / 111 (20%)
Selected to learn: Depo (Injection)	30 / 111 (27%)
Selected to learn: Traditional pill	11 / 111 (9.9%)
Selected to learn: Condoms	6 / 111 (5.4%)
Selected to learn: Copper IUD/coil	13 / 111 (12%)
Selected to learn: Rhythm/Cycle beads	10 / 111 (9.0%)
Selected to learn: Mini Pill	7 / 111 (6.3%)
Selected to learn: Hormonal IUD	3 / 111 (2.7%)
45 days postpartum	
Currently using FP	372 / 952 (39%)
Experienced FP side effects	112 / 312 (36%)
Shared reason for not using FP	512 / 2,201 (23%)
Reason: Do not need yet	63 / 512 (12%)
Reason: I am exclusively breastfeeding	112 / 512 (22%)
Reason: Do not want to	9 / 512 (1.8%)
Reason: Facility stocked out	13 / 512 (2.5%)
Reason: Worried about side effects	98 / 512 (19%)
Reason: Want more information before deciding	221 / 512 (43%)
2.5 months postpartum	
Considering introducing food to baby	306 / 932 (33%)
Has resumed menstruation	227 / 554 (41%)

Abbreviations: LAM = lactational amenorrhea method; FP = family planning; IUD = intrauterine device.

Notes: This table summarizes engagement of treatment group participants (N = 2,376) with follow-up counseling messages sent after delivery. At 21 days postpartum, women were invited to learn more about various topics including method safety, timing, and resuming sex; numerators and denominators in each row indicate women who provided a response to that specific prompt (percentages are calculated among respondents). Of those who engaged at 21 days, 36% requested more information about family planning methods. At 36 days, users were asked if they were still interested in their previously selected method and were invited to learn more; the most frequently selected methods were injectables (27%) and implants (20%). At 45 days postpartum, women were asked whether they had started a method; of those who responded, 39% said yes. Among those not using a method, the most common reasons were needing more information (43%), concerns about side effects (19%), and reliance on exclusive breastfeeding (22%). Messages at 2.5 months collected updates on menstruation and breastfeeding status to inform continued LAM use. Numerators and denominators reflect respondents to each question (percentages calculated among respondents).

Appendix Table A8. Method Mix

Characteristic	Overall N = 4,751	Control N = 2,374	Treatment N = 2,377
Prior Experience and Planned Use at Baseline			
Most recent method before study			
LAPMs	20%	20%	20%
SARC	38%	37%	38%
Barrier	2.6%	2.5%	2.7%
LAM	0%	0%	0%
Traditional	2.9%	2.9%	2.8%
No method	37%	37%	37%
Unsure/Refused	0.1%	0.1%	0.2%
Missing data	0%	0%	0%
Method in mind			
LAPMs	32%	31%	34%
SARC	26%	25%	27%
Barrier	0.3%	0.3%	0.2%
LAM	<0.1%	<0.1%	<0.1%
Traditional	0.7%	0.9%	0.5%
No method	15%	16%	15%
Unsure/Refused	25%	27%	24%
Missing data	0%	0%	0%
Contraceptive Uptake at Follow-up			
Method adopted at 3 months			
LAPMs	16%	16%	17%
SARC	29%	29%	29%
Barrier	0.6%	0.6%	0.6%
LAM	1.0%	0.7%	1.2%
Traditional	0.3%	0.2%	0.3%
No method	41%	41%	40%
Unsure/Refused	<0.1%	<0.1%	0.1%
Missing data	12%	12%	13%
Method adopted at 6 months			
LAPMs	23%	23%	24%
SARC	30%	31%	29%
Barrier	0.4%	0.4%	0.5%
LAM	0.2%	0.2%	0.1%
Traditional	0.4%	0.4%	0.3%
No method	25%	25%	24%
Unsure/Refused	<0.1%	<0.1%	0.1%
Missing data	21%	21%	22%

Abbreviations: LAPM = long-acting or permanent methods (sterilization, implants, IUDs); SARC = short-acting reversible contraception (injectables, pills); LAM = lactational amenorrhea method.

Notes: This table summarizes participants' reported contraceptive method use and preferences at four points: most recent method prior to study enrollment, postpartum method in mind at baseline, and methods adopted at 3 and 6 months postpartum. Categories include LAPMs, SARCs (injectables, pills), barrier methods, LAM, traditional methods (e.g., withdrawal, rhythm), no method, and "unsure/refused." Method in mind was assessed at baseline, while adoption was based on self-reported use at the 3- and 6-month follow-up surveys. Missing responses are indicated in the final row of each section. The sample is split by treatment arm and includes all randomized participants (N = 4,751).

Appendix Table A9. Summary of Outcomes Among Women in the Treatment Group by Engagement Type

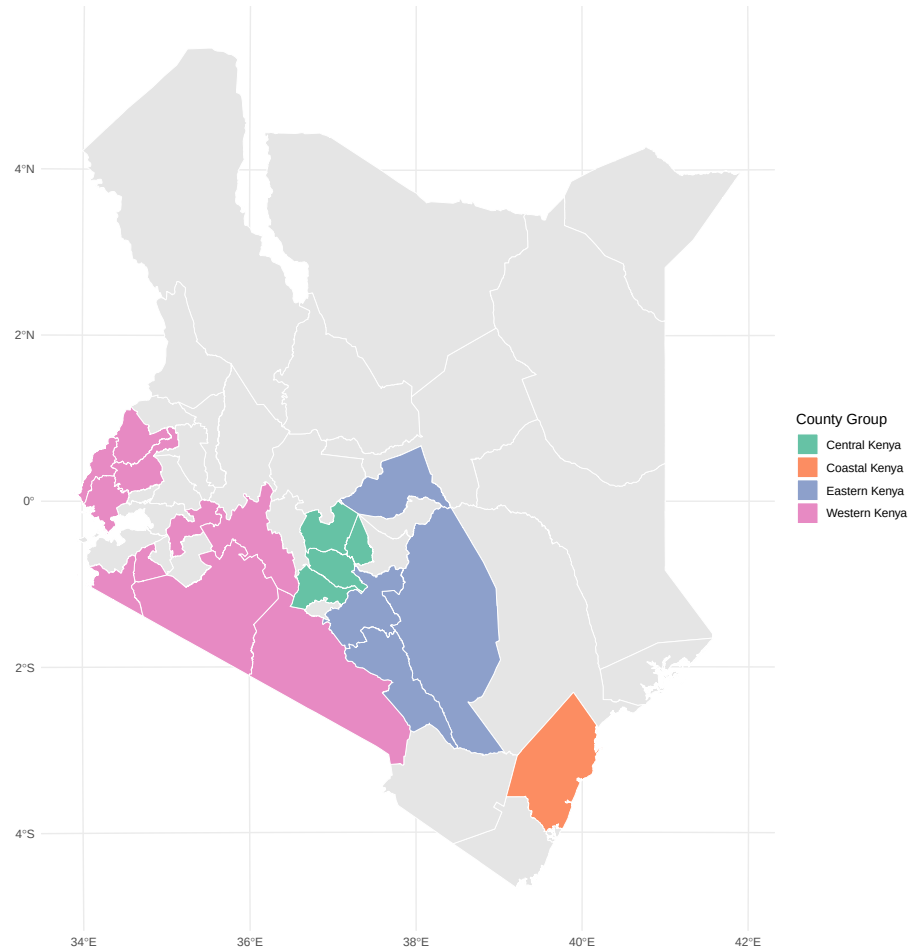
Characteristic	Non-SDM Participants N = 1,461	SDM Participants N = 377	p-value
Primary Outcomes			
Use of modern methods at 3 Months	749 / 1,458 (51%)	225 / 377 (60%)	0.004
Use of modern methods at 6 Months	881 / 1,293 (68%)	249 / 343 (73%)	0.11
Secondary Outcomes			
Knowledge of LAM at 3 Months	0.74 (0.86)	0.84 (0.86)	0.037
Use of LAPMs at 6 months	377 / 1,293 (29%)	121 / 343 (35%)	0.029
Method satisfaction at 6 Months	4.58 (1.01)	4.61 (0.95)	0.9
Intention to continue FP method at 6 months	699 / 845 (83%)	199 / 243 (82%)	0.8

Abbreviations: SDM = Shared Decision-Making; LAM = lactational amenorrhea method; FP = family planning.

Notes: This table compares outcomes between women in the treatment group who engaged with the SDM tool during pregnancy and were reached for follow-up (N = 377) and those in the treatment group who did not engage with the SDM or were not offered it (N = 1,950). SDM engagement refers to accepting the message prompt and participating in the preference-ranking flow. All outcomes are based on self-reports collected during the 3-month (midline) and 6-month (endline) surveys. Means and standard deviations (SD) are shown for continuous outcomes, and proportions for binary outcomes. P-values are from Wilcoxon rank-sum tests (continuous outcomes) or chi-square tests (binary outcomes).

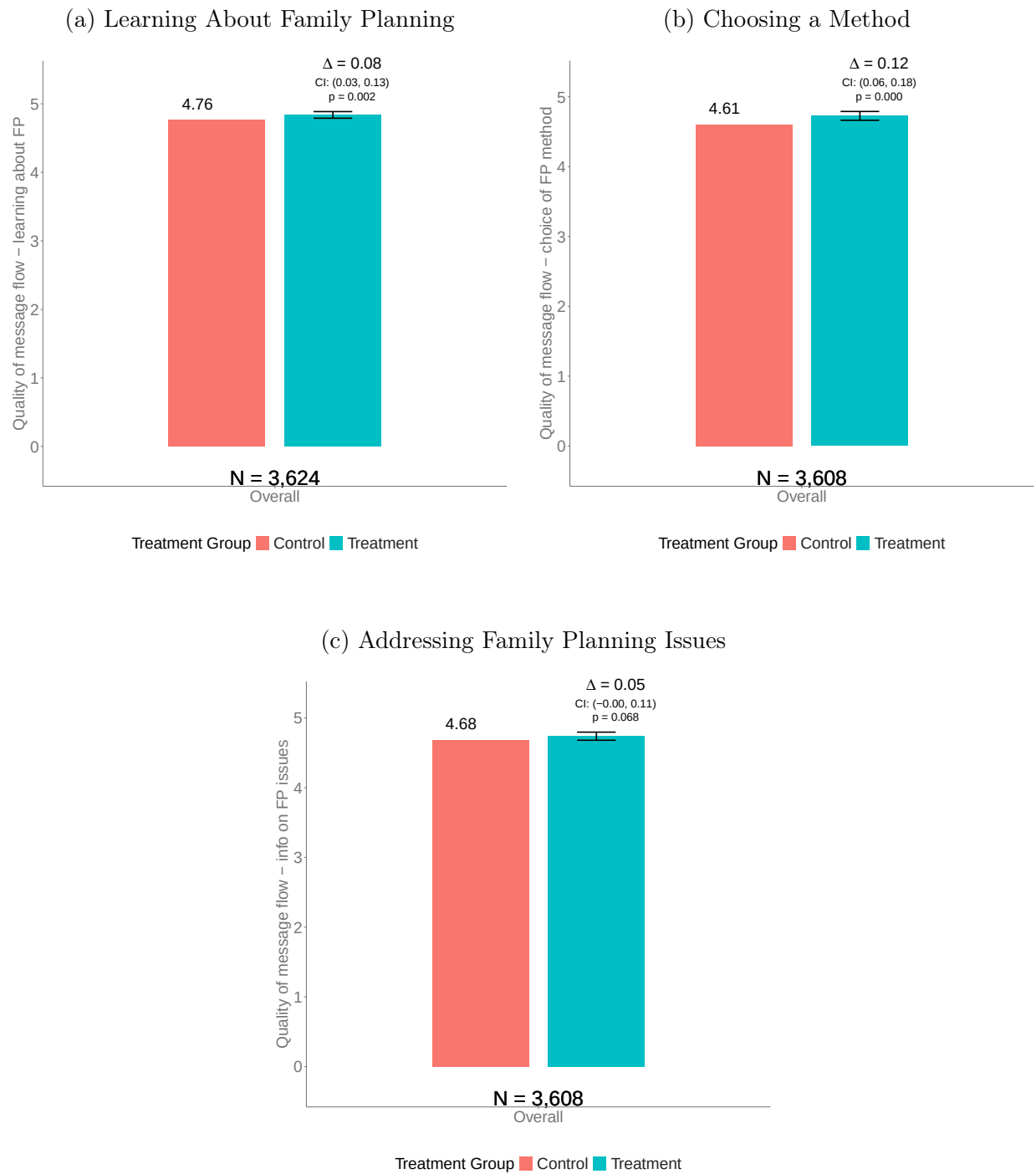
Appendix B: Figures

Figure B1. Geographical Distribution of the Study Sample



Notes: This map displays the geographical distribution of study participants across Kenya. The colored areas indicate the counties represented in the study sample, grouped into four major regions: Central, Coastal, Eastern, and Western Kenya.

Figure B2. User-Rated Quality of PROMPTS Message Flows (Sub-Items)



Notes: These figures present treatment effects on user-rated quality of the PROMPTS message flows related to family planning (FP), measured at the 6-month follow-up among women who confirmed receiving the messages. Participants were asked to rate how helpful the messages were in (a) learning about family planning methods, (b) choosing a method, and (c) providing useful information on FP issues. Ratings were collected on a scale from 1 (“not helpful at all”) to 5 (“very helpful”). The left bar in each figure shows the mean for the control group; the right bar shows the control group mean plus the estimated treatment effect. Confidence intervals and p-values are from linear regressions adjusting for stratification variables.

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